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**СИНТАКСИЧЕСКАЯ СЛОЖНОСТЬ ПИСЬМЕННЫХ ТЕКСТОВ
НА РУССКОМ И АНГЛИЙСКОМ ЯЗЫКАХ КАК ИНОСТРАННЫХ**

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**SYNTACTIC COMPLEXITY OF WRITTEN TEXTS
IN RUSSIAN AND ENGLISH AS FOREIGN LANGUAGES**

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Table of Contents

Abstract.....	4
Acknowledgements.....	4
1. Introduction.....	5
2. Related Work.....	7
3. Data.....	14
3.1. Corpora.....	14
3.1.1 CoRST.....	15
3.1.2 Pikabu.....	15
3.1.3. RLC.....	16
3.1.4. RULEC.....	16
3.1.5. ACTR.....	16
3.1.6. LOCNESS.....	17
3.1.7. Reddit.....	17
3.1.8. REALEC.....	17
3.2. Preprocessing.....	18
4. Syntactic Complexity Measures.....	21
4.1. Clause Extraction.....	21
4.2. T-unit Extraction.....	23
4.3. Coordination and Subordination.....	23
4.4. Clause Structure.....	24
4.5. Tree Depth.....	25
4.6. Mean Dependency Distance.....	25
4.7. Node-to-Terminal-Node Ratio.....	25
4.8. Nominal Phrase Extraction.....	25
5. Correlation of Complexity Measures.....	27
6. Relationship between Language, Proficiency, and Task Type.....	29
6.1. L2: Influence of Task Type.....	29
6.2. L2: Influence of Proficiency Level.....	30
6.3. L1: Influence of Language and Genre.....	34
6.3.1. Formal vs. Informal.....	34
6.3.2. English vs. Russian.....	36
6.4. Are Heritage and Foreign Texts different?.....	39
7. Predicting Proficiency.....	41
7.1. Classification.....	41
7.1.1. Logistic Regression.....	41
7.1.2. Decision Tree Classifier.....	43
7.1.3. Feature Importance.....	45
7.2. Regression.....	46
8. Future Work and Limitations.....	49
9. Conclusion.....	50

References.....	52
Appendices.....	59
Appendix 1. List of complexity measures, their abbreviations, and formulas.....	59
Appendix 2. Collinearity matrices (by language and speaker type).....	60
Appendix 3. Research repository on GitHub.....	62
Appendix 4. Python package for calculating syntactic complexity measures.....	62

Abstract

This study offers a comprehensive perspective of syntactic complexity in English and Russian texts written by L1 and L2 speakers. We analyze 20 syntactic complexity measures pertaining to the sentential, clausal, and phrasal levels, and explore their interrelationships, correlation with proficiency, task type, and genre. We propose a new measure of syntactic complexity based on Levenshtein distance at the clausal level. Our findings reveal strong correlations among length-based measures and highlight the problematic nature of the Coordination Index commonly used in the literature. We also find support for the idea that complexity generally increases with proficiency, with some measures plateauing at advanced levels. Syntactic complexity measures can also reliably distinguish between texts of different genres and task types; some values are language-specific, differing in the two languages considered.

Despite the challenging nature of our data, some complexity features, namely length-based indices and phrasal complexity measures, are useful in the task of automatic proficiency prediction. As a practical application of our research, we introduce *syntaxcomp*, a Python library for extracting syntactic complexity measures from CoNLL-U annotations.

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I am eternally grateful to my parents for the unconditional love and support they bestowed upon me throughout my university years. Without them, this work would not have been possible. Finally, I thank Irina Bagirokova, whom I was lucky enough to assist for the last five years, gaining invaluable first-hand experience in language teaching.

1. Introduction

Complexity has been studied extensively in many branches of linguistics, including but not limited to formal theoretical approaches, typology, experimental linguistics, and language acquisition (Housen et al., 2019). Both first and second language acquisition (L1A and SLA) researchers have taken a keen interest in the matter. One of the first endeavors into complexity as a factor in second language (L2) learning was made by Larsen-Freeman (1978), who posed the question of whether certain syntactic features of a text could serve as a “developmental yardstick” for determining a learner’s proficiency level.

In the following decades, many different types of complexity were studied and a vast number of measures (or *operationalizations*) were proposed. The types of complexity are named after the levels of language to which they pertain: phonological, morphological, lexical, and syntactic, with some researchers also distinguishing discursive complexity. In this study, we focus exclusively on the notion of syntactic complexity.

Despite a large number of publications and tools aimed at understanding syntactic complexity, still some questions remain unanswered. First of all, most of the existing research has focused on English as a second or foreign language, with only moderate attention being paid to other European languages and even less to non-European ones. Secondly, the effects of cross-linguistic influence of the learner’s L1 on the syntactic complexity of their L2 has not been examined sufficiently. Moreover, a vast majority of studies looking into syntactic complexity only deal with one type of speakers: either children acquiring their L1 or learners of a second or foreign language. In recent years, some progress was also made towards understanding the nature of syntactic complexity in heritage language (HL), with Kisselev and colleagues (2019; 2021; 2022) focusing on Russian heritage speakers. Finally, some methodological concerns have been raised by Ortega (2003), Norris and Ortega (2009), and Housen et al. (2019). They touch upon the lack of a generally accepted definition of (syntactic) complexity and the current approaches to operationalizing complexity.

Thus, we believe that a large-scale study into measures of syntactic complexity in SLA is warranted. In this paper, we aim to analyze various measures of complexity at the sentential, clausal, and phrasal levels and to investigate how they relate to each other and to language-external factors. Our research is carried out on written material of two languages, English and Russian, and of two speaker types, L1 and L2. We pose the following research questions (RQs):

- 1) How do various syntactic features of written texts correlate to each other?

- 2) How do syntactic features of L2 texts change with growing proficiency and how do they differ from L1 texts?
- 3) How does task type influence syntactic complexity?
- 4) Are texts of similar types in different languages similar or different with regard to their syntactic complexity?
- 5) Do texts in the same language differ depending on genre?
- 6) Which complexity measures are good predictors of a learner's proficiency?

The results of our study may prove useful in multiple domains. From the theoretical standpoint, we hope to shed more light on the nature of syntactic complexity in L2 and the interaction of various text complexity parameters. Practically, our research may help in solving several Natural Language Processing (NLP) tasks, such as automatic evaluation and proficiency level determination, as well as Grammatical Error Correction (GEC) and providing automatic feedback for learners and education workers. Finally, our insights might be useful as part of language teaching, highlighting the areas in which learners experience difficulties and the ways in which they differ from native speakers.

The rest of the paper is organized as follows. We start by reviewing the current state of the field and outlining the theoretical issues voiced in previous research (Section 2). In Section 3, we describe the corpora, datasets, and tools used in our study. We then discuss a number of existing complexity measures and suggest some modifications that may improve their reliability, along with describing our approach to their calculation (Section 4). The analysis of different measures and their relationships with each other and various other factors is described in Sections 5, 6, and 7, and we conclude by discussing the remaining issues and highlighting prospects for future work in Section 8.

2. Related Work

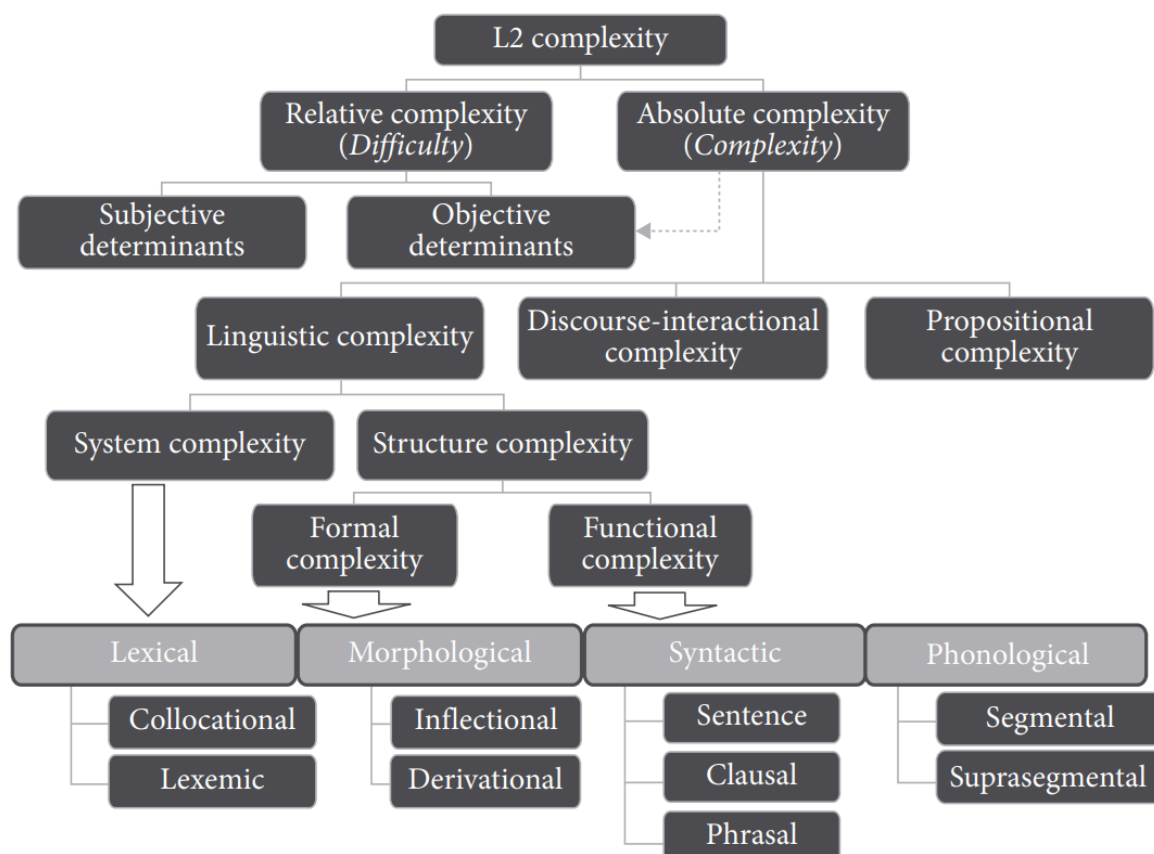
The notion of syntactic complexity has interested SLA researchers for several decades, starting at least as early as Larsen-Freeman (1978) who attempted to establish which syntactic features can be used as a yardstick for determining a learner's proficiency level. Among those features were, for example, "the percentage of error-free T-units and the average length of error-free T-units" (Larsen-Freeman, 1978: 439).

Despite a profound interest in the linguistic complexity phenomenon, to this day no generally accepted definition exists (Housen et al., 2019). In a broad sense, complexity (linguistic or otherwise) has been defined as "a matter of the quantity and variety of [a system's] constituent elements and of the interrelational elaborateness of their organizational and operational make-up" (Rescher, 1998: 1). In linguistics, it is customary to distinguish between absolute and relative complexity (Miestamo, 2008). Absolute complexity, also referred to as objective or structural complexity, reflects inherent properties of a language construction: for instance, one could say that a sentence containing two coordinate clauses has higher absolute complexity than a sentence consisting of only one clause. Absolute complexity is often estimated via length-based indices, such as the number of words per clause or clauses per sentence. On the other hand, relative (or subjective) complexity is tied to the difficulty of acquiring and/or processing a particular structure. It is thus closely linked with the concepts and ideas of experimental linguistics, and researchers often refer to it as user-based complexity (Laleko & Scontras, 2021) or difficulty (Housen & Simoens, 2016). In practice, it is usually measured by calculating ratios of simple vs. complex structures (Kisselev et al., 2021).

Encompassing both absolute and relative complexity, Ortega (2003: 492) defines syntactic complexity as "the range of forms that surface in language production and the degree of sophistication of such forms". This term is used interchangeably with linguistic complexity as a whole. On the other hand, Lu and Ai (2015) think of syntactic complexity as a part of linguistic complexity; their definition, however, is quite similar: "the range of *syntactic* structures that are produced and the degree of sophistication of those structures" (Lu & Ai, 2015: 17, emphasis ours). Similarly, Kisselev et al. (2021) consider syntactic complexity to be "only one aspect of linguistic complexity, which is distributed across all linguistic domains" (Kisselev et al., 2021: 2). Other types include phonological (Júnior, 2013), lexical (Skehan, 2003; Bulté et al., 2008), morphological (Verspoor et al., 2012; Brezina & Pallotti, 2016), and phraseological (Paquot & Granger, 2012; Paquot, 2017)

complexities, as well as discursive complexity (Panteleeva, 2019). Figure 1, taken from Bulté and Housen (2012), shows a hierarchy of complexity.

Figure 1. A taxonomy of complexity constructs (Bulté & Housen, 2012: 23).



It is important to distinguish between the notions of complexity, accuracy, and fluency (also known as CAF). Fluency can be defined as “the production of language in real time without undue pausing or hesitation” (Ellis & Barkhuizen, 2005: 139). Wolfe-Quintero et al. (1998) operationalize it as the amount of text a speaker produces in a given time period. Accuracy is perhaps self-explanatory and refers to “the degree of conformity to certain norms” (Pallotti, 2009), these norms being the rules of the target language grammar. It is measured as the degree or percentage of error-free language produced by a speaker (Wolfe-Quintero et al., 1998). Together, CAF are often used as indices of language development, even though it has been shown that accuracy by itself is not particularly suitable for the task of estimating development or proficiency (Bley-Vroman, 1983; Pallotti, 2009).

The research of linguistic complexity in the field of SLA is often based on the assumption that the complexity of a text increases along with the learner’s proficiency

(Ortega, 2012). By examining various complexity measures and their relationship with proficiency, one can single out metrics suitable for estimating a speaker's level of proficiency in the absence of an explicit indicator (a CEFR level, a grade, etc.). It could even be argued that complexity measures are more objective indices of proficiency, since they are less task- and examiner-dependent. Moreover, numeric complexity values allow for a straightforward comparison between studies, since they are calculated relatively on the same scale (unlike grades and proficiency levels, which there are multiple types of). Ortega (2012) criticizes the common practice of using categorical proficiency values to assess mean complexity differences between groups, since this approach leads to the loss of information: the decreased variance of proficiency scores may lead to a higher chance of type II errors (i.e., false negatives). She suggests that "both complexity and proficiency [be] treated as interval scales" (Ortega, 2012: 131).

Ortega (2003) conducted a research synthesis of 25 studies on the relationship between various syntactic measures and L2 English proficiency. Her focus was on the changes that occur in the process of language acquisition. She concluded that "syntactic complexity values varied systematically depending on instructional setting and proficiency sampling criterion" (Ortega, 2003: 512). Firstly, texts written by learners of English as a second language (SL) were characterized by a higher syntactic complexity than those of people learning English as a foreign language (FL). Secondly, comparing learners based on holistic ratings instead of programme level yielded more homogeneous complexity results. It was also found that a one-year observation period is preferable for a noticeable change in complexity values to occur.

However, Ortega (2003) notes that the relationship between (syntactic) complexity and proficiency is not linear. For example, in regard to subordination, more advanced learners are expected to display complexification at the phrasal (rather than clausal) level, as demonstrated e.g. by Monroe (1975) for French. This is supported by Halliday's (1998) Systemic Functional Linguistics (SFL) approach, which concentrates on language function in relation to social context. More specifically, SFL introduces a continuum of dynamic and synoptic styles of language, the former being associated with informal, spontaneous speech, and the latter with formal, carefully planned writing. Dynamic styles are characterized by "grammatical intricacy" (Vande Kopple, 1995: 3), namely a high amount of subordination. On the other hand, the main feature of synoptic styles is lexical density, i.e. the ratio of lexical (or content) words to the total number of words. Both in L1 and L2 acquisition, dynamic styles have been shown to emerge earlier than synoptic ones (e.g. Derewianka, 2003

for L1; Achugar & Colombi, 2008 for L2). In terms of SLA complexity, this explains why the amount of subordination is a useful indicator of complexification at intermediate levels of proficiency, but not so much when it comes to advanced learners. At this point, learners turn to grammatical metaphor processes, especially nominalization (Simon-Vandenberg et al., 2003). Therefore, for more advanced learners, another complexity measure is needed; Ortega (2003) thus proposes mean length of clause as an indicator of complexity that is typical of synoptic styles.

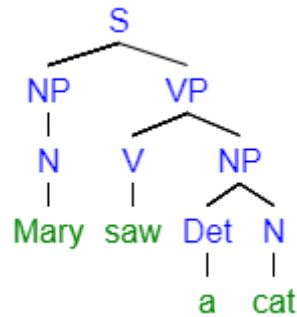
Importantly, proficiency is far from the only factor that can influence the characteristics of a learner's text. Ortega (2012) notes that many studies suffer from what she calls construct reductionism: the idea that the relationship between complexity and proficiency can be accurately modeled via approximations such as length (of a clause, T-unit, etc.), amount of subordination, and frequency of a chosen range of forms (usually those that occur rarely in native speech or are acquired late by L1 speakers). However, such factors as genre, task type, environment, and topic can also affect the outcome. It is therefore crucial to account for these variables in order to obtain a clear picture of the matter.

The literature describes a vast number of syntactic complexity measures, also called *operationalizations* (Szmrecsanyi, 2004), with studies reporting on dozens of various measures (Lu, 2011; Bulté & Housen, 2012; Verspoor et al., 2012, among others). Many are based on simple enumeration of certain elements (words, clauses, noun phrases, etc.), others involve ratios (words per clause, clauses per T-unit, and so on). Length-based indices such as the number of words, clauses, T-units, etc., as well as those measuring the amount of coordination and subordination, have received the most attention in the literature (Bulté & Housen, 2012). The notion of a T-unit, which is abbreviated from (*minimal*) *terminable unit*, was proposed by Hunt (1965) as “one main clause with all subordinate clauses attached to it” (Hunt, 1965: 20). In other words, T-units are “the shortest grammatically allowable sentences” into which an utterance can be split.

More complex measures also exist, for example, Szmrecsanyi's (2014) Index of Syntactic Complexity (ISC), which is computed by first counting the number of subordinating conjunctions, WH-pronouns, verb forms, and noun phrases (NPs) and then multiplying them by their weights (2 for conjunctions and pronouns, 1 for verbs and NPs). The author compares ISC to two other measures: word count and node count. The latter was proposed as a complexity measure in psycholinguistic studies carried out by Johnson (1966) and Ferreira (1991), and it represents the number of nodes in a constituency tree dominated

by the sentence. For instance, as illustrated by Figure 2, the sentence *Mary saw a cat* dominates 7 nodes (not counting the node S itself).

Figure 2. Constituency tree of the sentence *Mary saw a cat*.



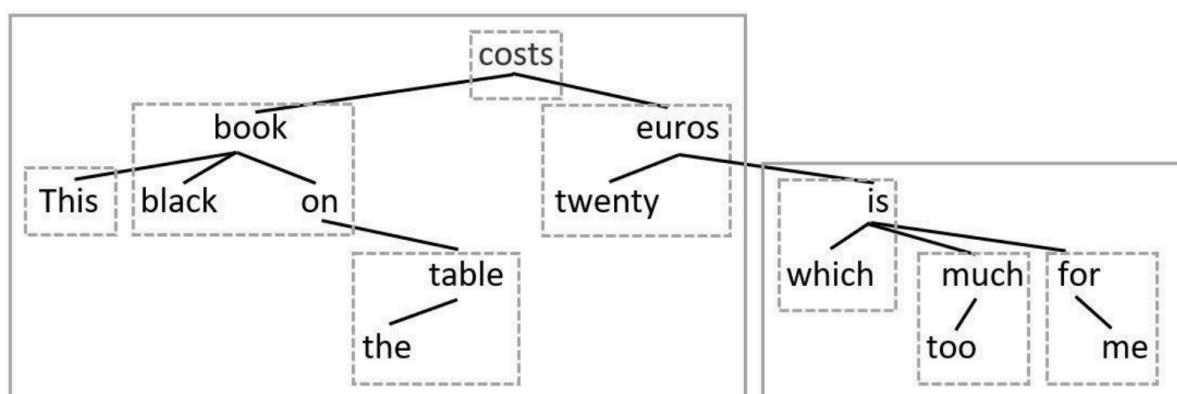
By computing Spearman’s correlation coefficient on the complexity rankings provided by the three measures, Szmrecsanyi (2014) found that they are highly correlated. The author thus concluded that ISL and simple word count are good proxies for node count, a measure he deemed “conceptually and psychologically the most real”, but computationally expensive.

Another measure of complexity at the phrasal level was explored by Panteleva (2019), who analyzed a variety of morphological, lexical, syntactic, and discursive features. The measure in question is dependency tree depth, i.e. the distance (number of nodes) “from the root of the tree to its farthest descendant” (Panteleva, 2019: 17). Average and maximum tree depth were found to be the strongest predictors of the grade received by an essay; however, minimum depth was statistically significant as well. A similar feature, nesting depth, was analyzed and found significant by Kisselev et al. (2021).

Dependency distance, i.e. the distance between two words in a sentence that are syntactically related, was originally introduced as a measure of cognitive difficulty in L1 research (Liu, 2008). However, it has also been successfully applied to L2 complexity; for example, Ouyang et al. (2022) investigated mean dependency distance (MDD) between subordinations, coordinations, noun modifiers, and overall, showing that the latter was significant for distinguishing between all pairs of adjacent proficiency levels.

More recently, a new syntactic complexity measure was proposed and tested on Czech material by Nogolová et al. (2023). They explored Linear Dependency Segments (LDSs), which were first described by Mačutek et al. (2021). An LDS is “the longest possible sequence of words (belonging to the same clause) in which all linear neighbours [...] are also syntactic neighbours” (Mačutek et al., 2021: Section 2).

Figure 3. LDS structure of the sentence *This black book on the table costs twenty euros, which is too much for me* (Mačutek et al., 2021). LDSs are enclosed in dotted rectangles.



The concept of an LDS is valuable because it reflects both dependency structure and word order. Nogolová et al. (2023) consider two measures: the average length of an LDS in words and the average length of a clause in LDSs. They find that the former decreased with growing proficiency, while the latter increased, bridging the gap between L2 and native speakers.

Until recently, the focus of SLA researchers studying linguistic complexity has been mostly on learners of English with an SL or FL background. Kisselev and Alsufieva (2017) broke ground by turning to heritage Russian speakers, outlining some differences in the way coordination and subordination developed in L2 and heritage learners' writing overtime. Kisselev et al. (2021) continue the research into the complexity of Russian as a heritage language (HL). They analyze a list of 12 variables previously applied to L2 data and conclude that most of these measures are applicable for differentiating between HL proficiency levels as well. Importantly, no two measures show a strong correlation (in fact, most of them are not correlated at all). The issue of carelessly choosing complexity measures that are highly correlated and essentially point to the same thing had been previously voiced by Norris and Ortega (2009) and Ortega (2012).

Kisselev et al. (2022) carried out a similar study on the material of the longitudinal corpus RULEC (see Section 3.1.4 for a detailed description) and reported largely similar findings; for instance, the proportion of coordinate clauses in FL learners' writing declined, while for HL learners it remained stable, with figures for both speaker types converging around the Advanced level. The authors conclude that the development of various syntactic features in learners' writing is non-linear and is intricately connected with genre, task type, and individual learner factors.

With the increase of computational capabilities and the widespread use of morphosyntactic parsers, more and more automatic tools for calculating measures of complexity emerge. For instance, Coh-Metrix (Graesser et al., 2004)¹, an online tool for measuring text cohesion, includes more than 100 parameters. Another example is the L2 Syntactic Complexity Analyzer (L2SCA; Lu, 2008)², which was recently tested on transcribed spoken English data (Kim & Lu, 2024). Panteleeva (2019) developed *Inspector*³, a tool offering comprehensive feedback to learners of Russian preparing for the International English Language Testing System (IELTS) exam. Inspector estimates a grade the learner is likely to receive, provides a comparison of the input text's values with other learners' essays, and suggests corrections. However, most of such tools were designed for the English language, leaving plenty of prospects for future research.

¹ <http://cohmetrix.memphis.edu/cohmetrixhome/>

² <https://sites.psu.edu/xxl13/l2sca/>

³ <https://github.com/panteleeva48/inspector>

3. Data

In order to study syntactic complexity in written L1 and L2 texts in Russian and English, a corpus consisting of several parts was collected. First, we included texts in both languages (English and Russian) and of both types (L1 and L2); second, since we aim to analyze the relationship between syntactic complexity and language proficiency, the L2 texts come from speakers of different proficiency. Moreover, to make our findings more sound and generalizable, we look at texts of different styles, e.g. formal and informal. This distinction is rather rough, and we use it mostly to distinguish between academic and non-academic texts. Overall, the data come from 8 different sources, adding up to 46 thousand texts (over 13 million tokens, excluding punctuation).

As noted by Ortega (2012), there are many language-external factors that can influence syntactic complexity. Therefore, we include the metadata provided by the corpora in our analysis. Unfortunately, each corpus contains a different set of meta-information, making a comparison across all corpora impossible. Additionally, some variables are encoded differently: for example, the proficiency level can be given in the form of a CEFR grade, hours of instruction, or a grade received for a specific task. The metadata present in each corpus are described in section 3.1.

3.1. Corpora

Table 1 shows a summary of the corpora used in our research. For L2 corpora, speaker types (foreign vs. heritage) are specified in brackets. More details on each corpus are given below.

Table 1. Summary statistics of the corpora used for research (after outlier removal).

Language	Speaker Type	Text Type	Corpus Name	Corpus Statistics				
				Texts	Sentences		Words ⁴	
				Total	Total	Mean per text (std)	Total	Mean per text (std)
Russian	L1	formal	CoRST	1,898	48,625	26 (19)	796,986	420 (301)
		informal	Pikabu	18,577	428,724	23 (17)	5,434,797	293 (194)

⁴ Punctuation is not included in the word count.

	L2	formal	RULEC (FL)	2,148	27,660	13 (9)	388,483	181 (123)
		informal	RLC (FL, HL)	1,297	21,404	17 (10)	263,730	203 (146)
			ACTR (FL, HL)	654	13,693	21 (9)	167,871	257 (122)
English	L1	formal	LOCNESS	393	13,789	35 (19)	275,690	702 (345)
		informal	Reddit	18,720	315,599	17 (12)	5,681,422	303 (198)
	L2	formal	REALEC (graph desc., FL)	1,255	11,621	9 (3)	226,391	180 (36)
		informal	REALEC (opinion essay, FL)	1,235	18,013	15 (4)	329,460	267 (54)
	Total				46,177	899,128	19 (15)	13,564,830

3.1.1 CoRST

For academic L1 Russian, we use the Corpus of Russian Student Texts (CoRST; Zevakhina & Dzhakupova, 2015), which is a collection of texts written by university students. As stated on the corpus webpage⁵, these texts are “course papers, term papers, bachelor’s and master’s theses, essays, abstracts, reports, summaries, autobiographies, and paragraphs (small texts for different purposes: home assignments, answers to various questions, etc.)” (Corpus of Russian Student Texts, n.d.). As of February 2024, CoRST contains 3,868 texts of varied length. For our research, we only use texts from categories which have a mean word count of 200-600 words per text in order to approximate the length distribution found in the other corpora. In addition, CoRST provides information on the author’s year of study, gender, and specialty.

3.1.2 Pikabu

The Pikabu dataset (Gusev, 2024)⁶ contains almost 7 million posts with comments from the pikabu.ru⁷ website, which is a Russian social networking service similar to Reddit. It allows users to post texts, images, videos, etc., while also being able to up- or downvote the

⁵ http://web-corpora.net/learner_corpus

⁶ <https://huggingface.co/datasets/IlyaGusev/pikabu>

⁷ <https://pikabu.ru/>

posts and leave comments. This dataset thus contains informal, non-academic texts in L1 Russian. For our purposes, we only use a subsample of 20,000 posts (not including comments) chosen to match the text length distribution of the other corpora (for a detailed description of the subsampling process, refer to Section 3.2). The metadata include the author's ID and username, the link to the post, its rating and tags, neither of which we include in our analysis.

3.1.3. *RLC*

Informal L2 Russian texts are taken from the Russian Learner Corpus (RLC; Rakhilina et al., 2016)⁸, which contains 2004 error-annotated texts written by speakers of Russian as a foreign or a heritage language. The texts are tagged with type (oral/written), genre, speaker type (FL/HL), the speaker's L1, proficiency in Russian, and gender.

3.1.4. *RULEC*

Formal L2 Russian texts are found in a subcorpus of RLC called RULEC – the Russian Learner Corpus of Academic Writing (Alsufieva et al., 2012)⁹. Unlike the other corpora used in this research, RULEC is a longitudinal corpus, containing texts written by 39 FL and HL speakers over a 4-year period of studying Russian at an American university. RULEC encompasses texts of several types: paragraph-long essays, one-sentence answers to a task prompt, lists of bullet points, etc. For our research, we only use paragraphs, since they are the most similar to texts found in the other corpora. RULEC contains individually written essays and some examples of group work; we exclude the latter since such texts may be written by learners of different proficiency, and it is unclear which proficiency level should be attributed to the text as a whole. Similarly to RLC, RULEC offers the following meta-information: the speaker's background (FL/HL), proficiency level (established through external testing), gender, and the text's function (argumentation, narration, persuasion, etc.).

3.1.5. *ACTR*

Along with RLC and RULEC, we use argumentative (informal) essays from the annual Russian National Post-Secondary Russian Essay Contest held by the American Council of Teachers (ACTR). These data are not publicly available, but have been used in previous research (for example, Kisselev, 2019)¹⁰. Same as with RULEC, we exclude texts written by a group of students. ACTR provides information on the speaker's background (FL/HL),

⁸ <http://web-corpora.net/RLC/>

⁹ <http://www.web-corpora.net/RLC/rulec>

¹⁰ We thank Olesya Kisselev for providing us with access to the dataset.

proficiency level, gender, age, home country, native language, and institution; however, a lot of the values are missing, with only background and proficiency being consistently documented across all texts.

3.1.6. *LOCNESS*

For academic L1 English, we turn to the Louvain Corpus of Native English Essays (LOCNESS; Granger, 1998) which contains 412 academic texts. LOCNESS is publicly available for research purposes. As stated on the corpus webpage¹¹, it includes British and American university students' essays, as well as British pupils' A-level essays. In addition to level (university/A-level), the corpus offers information on essay type (argumentative, literary, expository-historical, or mixed).

3.1.7. *Reddit*

Informal L1 English texts are taken from the Webis-TLDR-17 Corpus (Syed et al., 2017)¹², which contains almost 4 million posts and replies from the American social network Reddit. Since the size of this corpus is much larger than that of the L2 corpora, we only use original posts (not comments). As with Pikabu, we further downsize the corpus by randomly sampling 20,000 posts. The Webis-TLDR-17 Corpus was originally designed for abstractive summarization tasks, and it contains texts along with their summaries written under the tag TL;DR¹³. We do not use these summaries in our research.

3.1.8. *REALEC*

Finally, for L2 English, both formal and informal texts can be found in the Russian Error-Annotated Learner English Corpus (REALEC; Kuzmenko & Kutuzov, 2014)¹⁴. REALEC contains texts of two types: graph descriptions and argumentative essays, written in accordance with IELTS writing test requirements. We consider the former formal and the latter informal. In total, the corpus contains 32,009 texts; however, most of them are not explicitly marked with type (graph description, i.e. Type 1, or opinion essay, i.e. Type 2) or do not contain a mark or a CEFR level. Thus, we only use the part that has the necessary metadata, namely 1,255 graph descriptions and 1,235 argumentative essays. In our analysis, we treat these two groups as separate subcorpora, since they differ both in genre and in length (the recommended word count is 150 for a graph description and 250 for an argumentative essay).

¹¹ <https://uclouvain.be/en/research-institutes/ilc/cecl/locness.html>

¹² <https://zenodo.org/records/1043504>

¹³ "Too long; didn't read".

¹⁴ <https://realec.org/>

3.2. Preprocessing

The first preprocessing step was the unification of metadata within each corpora, since in some cases the information was manually encoded into the texts themselves or documented inconsistently by different annotators. The metadata were extracted via regular expressions (Python’s *re* package¹⁵) and converted to a single format for ease of analysis. Unfortunately, as mentioned above, all the corpora have a different approach to encoding metadata, so complete unification was not possible.

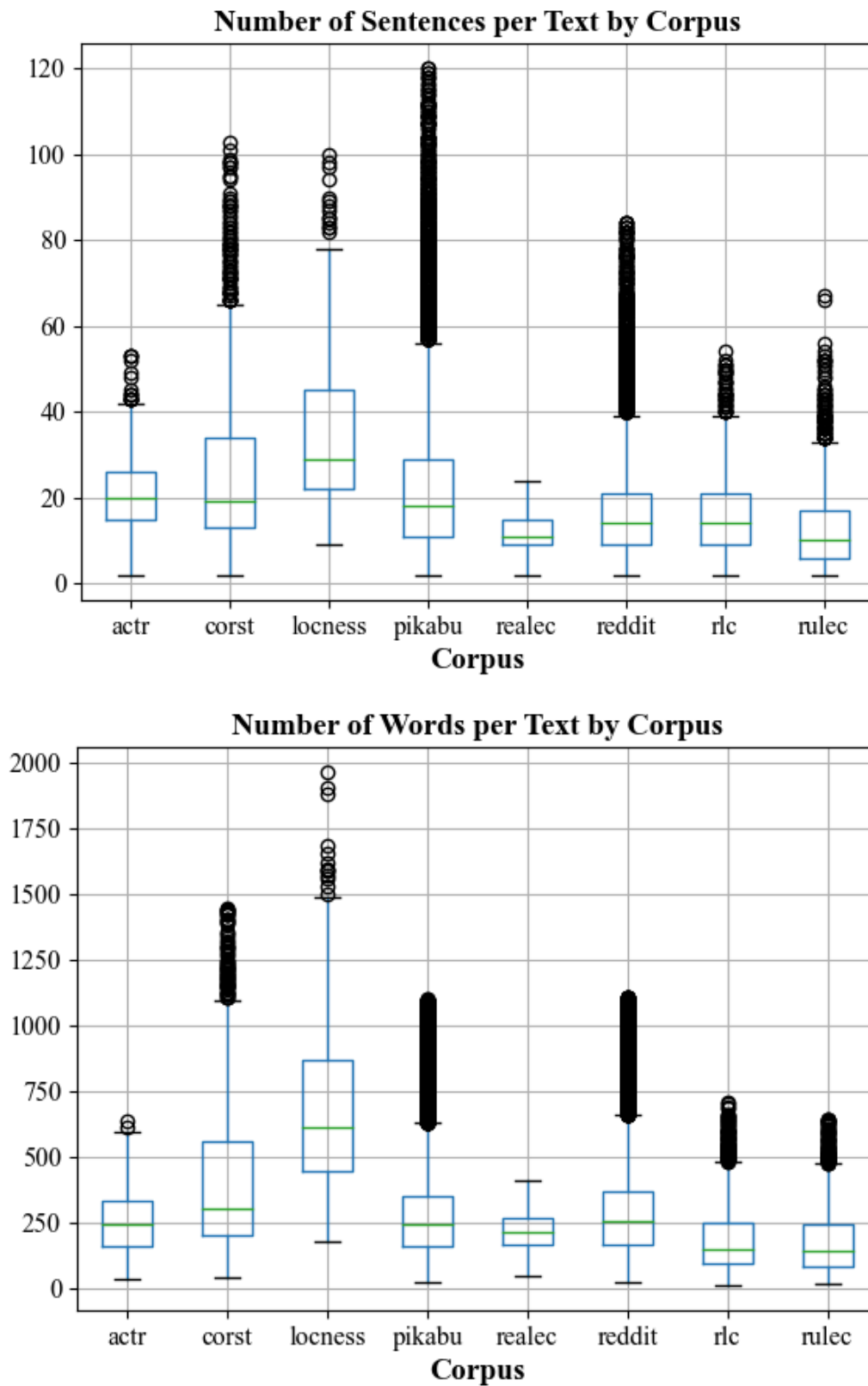
Due to the large size of the Pikabu and Reddit datasets, we employed a sub-sampling technique based on the length distribution of the other six corpora. We combined all texts from the smaller corpora into groups based on length and calculated the number of texts in each category. After that, 20,000 texts were randomly sampled from both Pikabu and Reddit based on class weights (i.e., the number of texts in each group). For all eight corpora, outliers by number of words and number of sentences were removed using groupwise z-scores. Additionally, after manual inspection, we noticed that some texts had been incorrectly perceived by the UDPipe parser as single long sentences, even though they contained terminal punctuation marks. These texts mainly came from the largest two datasets, Pikabu ($n = 225$) and Reddit ($n = 114$). To ensure that our syntactic measures were not skewed by these errors, we filtered out texts that were tagged as having one sentence but contained a period, an exclamation point, a question mark, or an ellipsis. No single-sentence texts remained in the corpus after the exclusion of such cases.

In L2 complexity research, it is customary to exclude short texts (less than 50 words) from consideration (Kisselev & Alsufieva, 2017; Kisselev et al., 2022; Nogolová et al., 2023). The reason for this is that using short texts for automatic analysis is methodologically questionable, since measures calculated on the basis of one or two sentences hardly present a reliable source of information. Nogolová et al. (2023: 194) set their threshold at 55 words, because “the standard-length requirement for passing a written exam is typically around 50–60 words”. We applied a less conservative threshold, namely 45 words and 5 sentences for each text. The motivation for this is as follows: short texts that include 5 or more sentences may simply be the production of less experienced, low-proficiency writers. Since there are few of those in our dataset, it is crucial not to exclude them. On the other hand, 2–4-sentence texts of less than 45 words turned out to be either brief answers to test questions or simply texts that were mistakenly cut off during corpus collection. Thus, our threshold was

¹⁵ <https://docs.python.org/3/library/re.html>

chosen on the basis of manual inspection. Figure 4 shows distribution of text length by corpus.

Figure 4. Length distribution of texts (in sentences and words) by corpus.



Following Panteleva (2019), we add a spell-checking step before morphosyntactic annotation. As shown by Lyashevskaya and Panteleva (2018), spelling errors negatively influence parsing quality. To avoid this, we employ Yandex Speller¹⁶ through the Python package *pyaspeller*¹⁷. Yandex Speller offers a CatBoost-based spell-checker capable of taking context into account. It is available for Russian, English, and Ukrainian, and offers a good trade-off between accuracy and computational efficacy (in comparison to simpler tools, which only work with one word at a time, or transformer-based models, which require a lot of computational power and are rather slow). Nevertheless, errors such as wrong or unnecessary corrections are possible, since no automatic spell-checking tool is 100% accurate. After carefully examining the results of spell-checking, we came to the conclusion that this step is mostly beneficial for L2 texts, but not for L1, since spelling errors in those are few.

Some texts in the ACTR corpus were written with a mixture of cyrillic and latin characters, because the corpus creators tried to preserve as much of the original spelling as possible while digitizing handwritten essays. Most of these cases were tagged and stored separately from the text itself, and the spelling for standardized (Kisselev et al., 2021), but for those that were not, we also added a transliteration step via the *transliterate* package for Python¹⁸.

Finally, for morphosyntactic annotation, we turn to UDPipe 2, specifically the models based on Universal Dependencies (UD) 2.12 treebanks with the addition of multilingual BERT (Straka & Straková, 2020). We use the default models: *english-ewt-ud-2.12-230717* for English and *russian-syntagrus-ud-2.12-230717* for Russian. UDPipe 2.12 models are available as an online service or via REST API¹⁹. The annotations can be retrieved in CoNLL-U format²⁰, which includes the token’s ID (i.e. position in the sentence), form, lemma, part-of-speech, morphological features, as well as the index of its head and the dependency relationship to it.

The subsequent work on extracting various complexity measures was done using Python’s *conllu* package²¹. Section 4 provides detailed information on this process.

¹⁶ <https://yandex.ru/dev/speller/>

¹⁷ <https://pypi.org/project/pyaspeller/>

¹⁸ <https://pypi.org/project/transliterate/>

¹⁹ <https://lindat.mff.cuni.cz/services/udpipe/>

²⁰ <https://universaldependencies.org/format.html>

²¹ <https://pypi.org/project/conllu/>

4. Syntactic Complexity Measures

We start by calculating simple measures, such as number of words (excluding punctuation), number of sentences, etc. We then move on to extracting clauses and T-units, computing sentence tree depth, and relative measures, e.g., average sentence length in words, clauses, and T-units, average clause and T-unit length in words, etc. The full list of measures along with their formulas and abbreviations can be found in Appendix 1. We make all code necessary for their calculation available in the form of a Python package called *syntaxcomp*²². Additional code and materials used for analysis are published in our GitHub repository²³.

4.1. Clause Extraction

As noted by Ortega (2003), the definitions of many concepts used in SLA complexity research can be inconsistent across studies. This is especially true for clauses (for a discussion of the concept of clause and clausehood tests, refer to Haspelmath (2016), section 2.2). In our work, we follow the approach described in the UD annotation guidelines, which are available at the UD website²⁴ (see also de Marneffe et al., 2014; Nivre et al., 2016; and de Marneffe & Manning, 2016). In short,

[a] simple clause minimally consists of a predicate together with its core argument dependents, but may be extended with oblique modifiers. [...] In complex clauses, both core arguments and oblique modifiers can also be realized as subordinate clauses. (Universal Dependencies guidelines, n.d.: Simple Clauses)

To extract clauses, we adopt an approach similar to that of Fan et al. (2013) who retrieve clauses from constituency trees and syntactic dependency trees. Given that UDPipe only provides information on dependency relationships and not constituents, we rely on the second method, albeit using constituents is somewhat easier.

Using the *conllu* package, each sentence is represented as a flat list of tokens and as a dependency tree. First, heads of clauses are identified through the *deprel* feature. The list of relations that indicate which tokens govern clauses was compiled on the basis of the information from the Universal Dependencies website, which in turn is a revised version of de Marneffe et al. (2014). The following relations are taken into account (definitions are retrieved from the UD website page pertaining to the *deprel* feature²⁵):

- *root*: root of the sentence
- *acl*: clausal modifier of a noun (adnominal clause)
- *acl:relcl*: relative clause modifier

²² <https://pypi.org/project/syntaxcomp/>

²³ https://github.com/eaklykova/syntactic_complexity

²⁴ <https://universaldependencies.org/guidelines.html>

²⁵ <https://universaldependencies.org/u/dep/index.html>

- *advcl*: adverbial clause modifier
- *advcl:relcl*: adverbial relative clause modifier
- *ccomp*: clausal complement
- *csubj*: clausal subject
- *csubj:outer*: outer clause clausal subject
- *nsubj:outer*: outer clause nominal subject
- *parataxis*: parataxis
- *xcomp*: open clausal complement (only for tokens with *upos* = VERB)
- *conj*: conjunct (only for tokens with *upos* = VERB)

Next, for each head, all its descendants are identified by recursively iterating over the head's children, grandchildren, etc. If a descendant is encountered that itself governs another clause, it and its descendants are marked as a new clause. Punctuation and other symbols (*upos* tags PUNCT and SYM) are excluded. The output is a list of dictionaries, where each element represents a clause and stores information on the head, dependents, clause type (marked by the *deprel* feature), as well as the length and text of the clause. Figure 5 shows the dependency tree of sentence (1) in the form of the *conllu* package output. Key features are highlighted in yellow, and elements that are excluded are marked with gray. Green frames denote separate clauses.

(1) *What is wrong, and what can we do about it?*

Figure 5. Syntactic dependency tree of a sentence with two clauses.

```
(deprel:root) form:wrong lemma:wrong upos:ADJ [3]
  (deprel:nsubj) form:What lemma:what upos:PRON [1]
  (deprel:cop) form:is lemma:be upos:AUX [2]

(deprel:conj) form:do lemma:do upos:VERB [9]
  (deprel:punct) form:, lemma:, upos:PUNCT [4]
  (deprel:cc) form:and lemma:and upos:CCONJ [5]
  (deprel:obj) form:what lemma:what upos:PRON [6]
  (deprel:aux) form:can lemma:can upos:AUX [7]
  (deprel:nsubj) form:we lemma:we upos:PRON [8]
  (deprel:obl) form:it lemma:it upos:PRON [11]
    (deprel:case) form:about lemma:about upos:ADP [10]
  (deprel:punct) form:? lemma:? upos:PUNCT [12]
```

The tree above is divided into two clauses: (1) *what is wrong*; (2) *and what can we do about it*.

4.2. T-unit Extraction

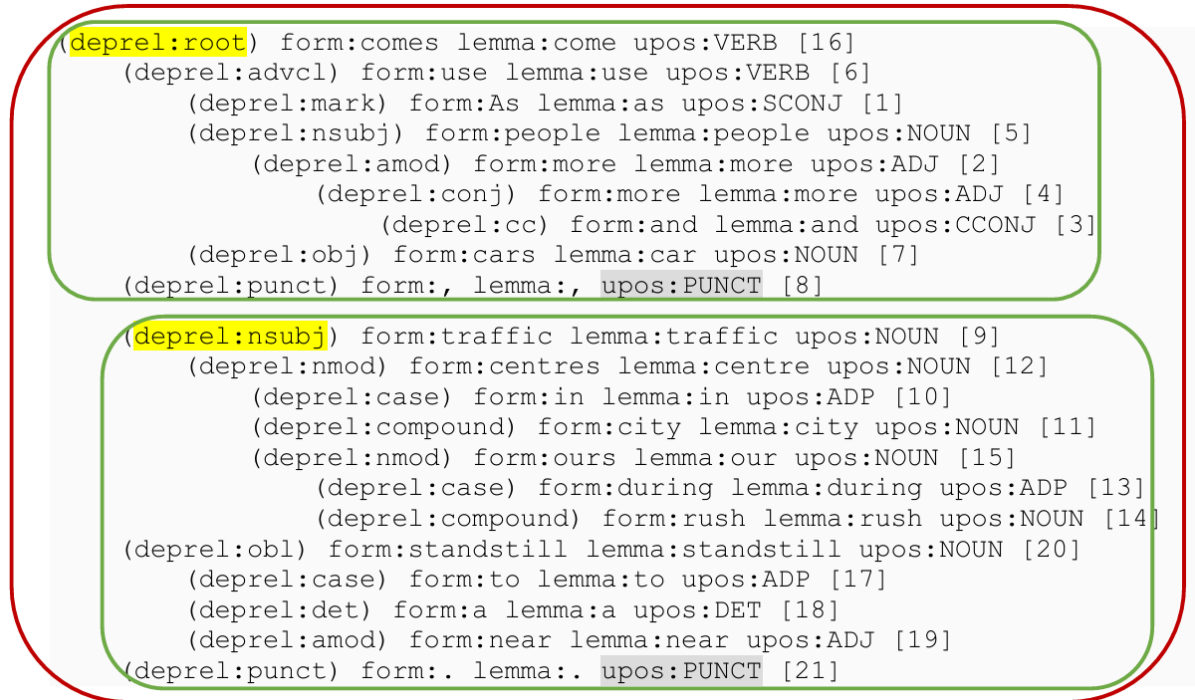
The dependency relationships suitable for extracting T-units are *root*, *conj* (for verbs) and *parataxis*, which is “a discourse-like equivalent of coordination” (Universal Dependencies, n.d.). The process is otherwise similar to that of identifying clauses.

In sentence (1), both clauses are simultaneously T-units; in (2), however, there are two clauses but only one T-unit:

- (2) *As more and more people use cars, traffic in city centres during rush hours comes to a near standstill.*

Here, the T-unit coincides with the sentence as a whole (see Figure 6, where it is framed in red). This is consistent with Hunt’s (1965) notion of a T-unit, since “[e]ach [T-]unit is grammatically capable of being considered a sentence” (Hunt, 1965: 21) and, in practice, they are often one and the same. Clauses, on the other hand, do not necessarily have the property of being independently grammatical; for example, the first clause of sentence (2) *As more and more people use cars* cannot be used on its own.

Figure 6. Syntactic dependency tree of a sentence with two clauses and one T-unit.



4.3. Coordination and Subordination

Following the recommendations of Norris and Ortega (2009), it is customary to estimate the degree of coordination in a text by calculating the proportion of coordinate clauses (see, for example, Kisselev et al., 2022). This is often referred to as the Coordination

Index (CI), proposed by Bardovi-Harlig (1992) as the ratio of coordinate clauses to all combined clauses, which are calculated by subtracting the number of sentences from the total number of clauses (since every sentence must have one independent clause). We include this measure in our analysis along with its equivalent for subordinate clauses.

However, Bulté and Housen (2012) criticize the CI because it is based on the total number of combined clauses (including subordinate ones); hence, if the number of coordinated clauses in the sentence remains equal and a subordinate clause is added, the CI value decreases. They believe that this measure “cannot be viewed as a ‘pure’ measure of clause coordination [...], since, ultimately, [it] depends on the amount of subordination produced” (Bulté & Housen, 2012: 38). They argue for the use of a different measure: the number of coordinate clauses per sentence, which is insensitive to the amount of subordination. In order to test the relationship between various measures of subordination and coordination, we include this feature as well.

Finally, we calculate the coordinate-to-subordinate clause ratio, because it can be useful for detecting overt shifts from coordination to subordination.

4.4. Clause Structure

Following the creators of Coh-Metrix McNamara et al. (2014), Panteleeva (2019) used Levenshtein distance to determine the similarity of sentences in the text. She calculated the mean pairwise distance between chains of lemmas and part-of-speech tags, as well as the mean distance between adjacent sentences. These features turned out statistically significant for distinguishing between best and non-best essays in the REALEC corpus²⁶ according to analysis-of-variance (ANOVA) tests.

We find that this measure can be further refined by transferring it to the clausal instead of the sentential level. It is unlikely that the learners will use identically structured multi-clausal sentences, but similarities are possible at the level of individual clauses. Moreover, instead of lemmas, we suggest using chains of dependency relationships (the *deprel* feature) in addition to POS-tags. The motivation for this is that dependency relationships may reveal tendencies that are obscured when only looking at parts of speech: for instance, both nominal and pronominal subjects are tagged with the *nsubj* relationship, while their parts of speech are different.

The Levenshtein distance, named after Levenshtein (1965), is the minimum number of edits needed to transform one word into another (the possible edits being deletion, insertion,

²⁶ The “best” essays were the ones graded 7 or above.

and substitution). For lists, every element is viewed as one symbol and the whole sequence as a word. We use the *textdistance* Python package²⁷ to calculate the pairwise distance between all clauses in a text and then take the average.

4.5. Tree Depth

As proposed by Panteleeva (2018; 2019), we calculate mean, maximum, and minimum dependency tree depth, all of which were found significant in distinguishing between best and non-best essays. However, we propose a slight change based on Kisselev et al. (2022) who suggest that this measure can be sensitive to errors of parsing or memorization of certain text fragments by learners. Kisselev et al. (2022) deal with this problem by calculating maximum nesting depth based on depth values that are encountered in the text at least three times. This might not work for shorter texts. Our idea is to use median tree depth instead of (or in addition to) mean, since the median is less sensitive to outliers.

4.6. Mean Dependency Distance

Following Ouyang et al. (2022), we take into account mean dependency distance. MDD is calculated as the average of all distances between a token and its head. We obtain these distances simply by subtracting the token’s ID (index in the sentence) from that of its head and finding the absolute value of the difference.

4.7. Node-to-Terminal-Node Ratio

This measure was suggested by Miller and Chomsky (1963) as a “rough measure of structural complexity”, which reflects “the amount of computation per input symbol that must be performed by the listener” (Miller & Chomsky, 1983: 480). We implement this measure by iterating over all nodes in the text sentence-by-sentence and separating nodes that have dependents from those that do not. After that, the total number of nodes is divided by that of the terminal nodes.

4.8. Nominal Phrase Extraction

Norris and Ortega (2009), followed by Bulté and Housen (2012), argue that greater attention should be paid to measures of complexity at the phrasal level. This can be achieved, for instance, by calculating the number of complex nominals per clause (Kisselev et al., 2022) or the mean length of noun phrases (Bulté and Housen, 2014).

²⁷ <https://pypi.org/project/textdistance/>

We rely on the former and operationalize it on the basis of the definition of nominal phrases offered by UD²⁸. An NP must have at least one (proper) noun or pronoun, which is the head, and may include the following modifiers (we only consider constructions that are possible in English or Russian):

- *nmod*: nominal modifier
- *nmod:poss*: possessive nominal modifier
- *nmod:tmod*: temporal modifier
- *appos*: appositional modifier
- *amod*: adjectival modifier
- *nummod*: numeric modifier
- *nummod:gov*: numeric modifier governing the case of the noun
- *det*: determiner
- *case*: case marking (for prepositions)

Note that we do not include clausal modifiers marked by *acl* and *acl:relcl*. This decision is motivated by our aim to distinguish phrase- and sentence-level complexity measures as much as possible.

The process of identifying these nominal phrases is widely similar to the one applied to clauses and T-units, with one notable exception: since NPs can be nested, we added a step to check that the head of an NP was not the dependent inside another NP. After identifying all NPs, we calculate the mean length of NP per text and the ratio of NPs consisting of two or more words.

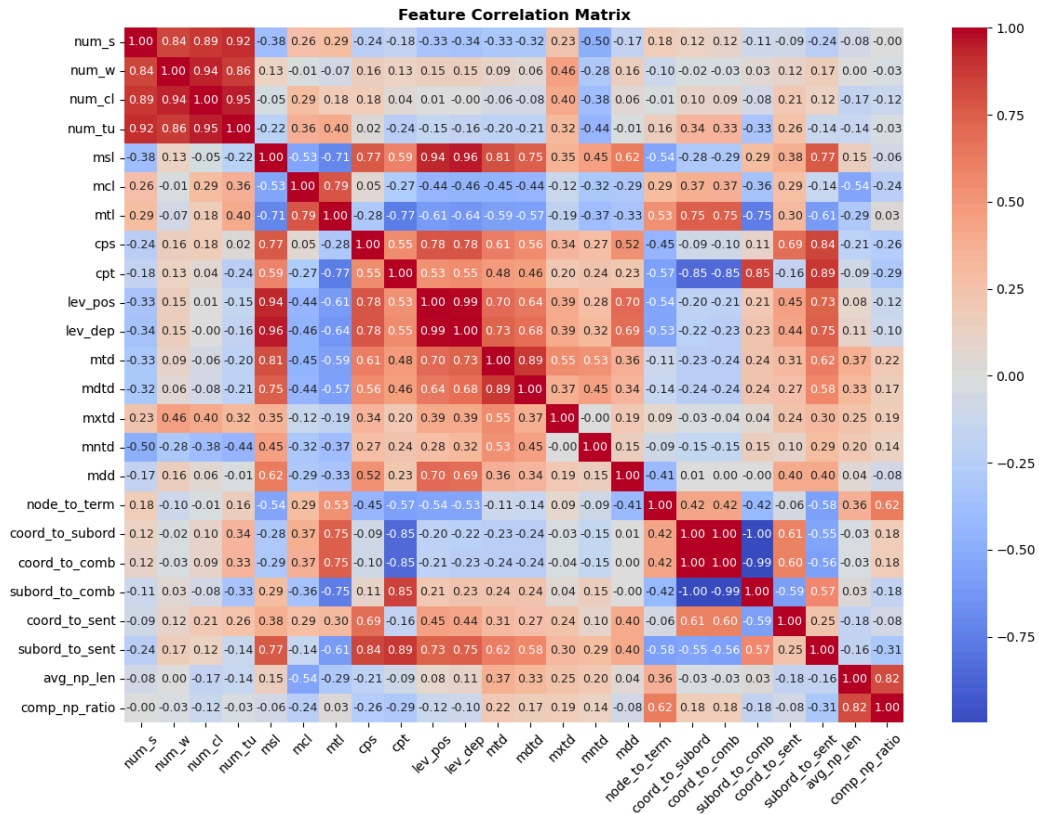
²⁸ <https://universaldependencies.org/u/overview/nominal-syntax.html>

5. Correlation of Complexity Measures

Let us start by taking a look at the correlations between measures. First, we compute a collinearity matrix for the whole dataset (both languages and both speaker types), and then move on to analyzing its various parts.

To choose a suitable correlation coefficient, we used the Shapiro-Wilk test to determine whether our values were normally distributed. The test revealed that none of the values conformed to the normal distribution, with p-values too small to display. Therefore, we used Spearman's rank coefficient, the values of which are presented in Figure 7.

Figure 7. Correlation matrix of syntactic complexity measures.



Perhaps unsurprisingly, the absolute values (number of words, sentences, clauses, and T-units) are all highly correlated. The two types of Levenshtein distance between clauses are closely linked with a Spearman rank coefficient of 0.99. The same can be said for mean and median tree depth, which have a coefficient of 0.89.

Mean sentence length (*msl*) is most strongly correlated with Levenshtein distance between clauses, which also seems natural: the more clauses in text, the longer the sentences, the bigger the mean difference between clauses (since there are simply more of them). This is supported by the positive correlation between the number of clauses per sentence (*cps*) and *msl*, as well as between *msl* and the ratio of subordinate clauses to sentences (the more

subordinates in a text, the longer the sentences). Mean and median tree depth also increase with increasing *msl*.

The coordinate-to-combined-clause ratio and the subordinate-to-combined-clause ratio have a perfect negative correlation. This provides strong support for Bulté and Housen’s (2012) criticism of the Coordination Index described in Section 4.3 and suggests that these measures should be used with caution.

Regarding the amount of subordination, one can see that the ratio of subordinate clauses to sentences (*subord_to_sent*) has a strong positive correlation with the number of clauses per sentence (*cps*) and per T-unit (*cpt*). Importantly, the same ratio for coordinate clauses has a weaker correlation to *cps* and a very weak negative relationship with *cpt*. This is consistent with the concept of a T-unit, which includes subordinate, but not coordinate clauses. The more subordinate clauses there are in a text, the more clauses are embedded into T-units, but the same does not hold for coordinates. On the other hand, the ratios of coordinate clauses to subordinate and all combined ones are strongly negatively correlated with *cpt*, – a fact that is also unsurprising since coordinate clauses constitute separate T-units.

The proportion of complex NPs is highly correlated with the average length of NP and has a less prominent, but still strong correlation with the Node-to-Terminal-Node Ratio. What this suggests is that the more nested the NPs are, the less terminal nodes there are: a node that could have been a leaf of the dependency tree becomes embedded into the complex noun phrase.

The positive relationship between the degree of subordination in the text (as illustrated by the *subord_to_sent* ratio) and the Levenshtein distances between clauses may be taken to mean that the more subordination there is in a text, the more diverse the clauses are structurally.

There were no notable differences between the correlation matrices computed for the English and Russian languages separately: the values were not exactly the same, but similar enough for all the tendencies described above to hold. The same is also true for texts separated by speaker type (L1 vs. L2). When comparing the correlation matrices for each of the four language-speaker type pairs (L1 Russian, L1 English, L2 Russian, L2 English), none of the patterns described above seem to be violated. The only difference is that the correlations seem to be the strongest for L2 Russian and the weakest for L2 English (see Appendix 2).

Given the multicollinearity of the investigated complexity measures, we return to the question of feature importance in Section 7.

6. Relationship between Language, Proficiency, and Task Type

In this section, we employ descriptive statistics and analyses of variance (ANOVAs) to uncover differences in syntactic complexity values based on language, proficiency level, and type of task. To run ANOVAs, we use the *stats* module of the *scipy* package for Python²⁹.

6.1. L2: Influence of Task Type

Following Panteleva (2019), we compare the two types of texts available in REALEC in terms of syntactic complexity. To test whether the differences in mean values are significant, we apply one-way ANOVAs with task type as the independent variable. All features turned out to be significant; average Levenshtein distance for dependency relationships was significant with a p-value smaller than 0.05, while for the other measures the p-value was under 0.001. This is consistent with Ortega's (2003; 2012) findings that task type and other factors related to instructional setting may influence text complexity. Table 2 presents the mean complexity values for the two tasks.

Table 2. Mean complexity values for two types of texts in REALEC and ANOVA results.

Complexity Measure	Mean (std)		F-statistic	p-value	Significance Level
	Graph description	Opinion essay			
<i>msl</i>	20 (5)	19 (4)	76,9	0	***
<i>mcl</i>	10 (2)	7 (1,02)	2680,1	0	***
<i>mtl</i>	16 (4)	14 (3)	316,6	0	***
<i>cps</i>	2 (0,56)	3 (0,62)	977,4	0	***
<i>cpt</i>	1,64 (0,39)	2 (0,36)	777,7	0	***
<i>lev_pos</i>	17 (4)	16 (4)	31,5	0	***
<i>lev_dep</i>	19 (5)	18 (4)	4,8	0,02899	*
<i>mtd</i>	6 (0,7)	5 (0,6)	56	0	***
<i>mdtd</i>	5 (0,82)	5 (0,71)	54,3	0	***
<i>mxttd</i>	8 (1,19)	8 (1,33)	51,6	0	***
<i>mntd</i>	4 (0,89)	3 (0,77)	247,8	0	***
<i>mdd</i>	3 (0,28)	3 (0,23)	153,8	0	***

²⁹ <https://pypi.org/project/scipy/>

<i>node_to_term</i>	0,81 (0,09)	0,73 (0,06)	673,2	0	***
<i>coord_to_subord</i>	0,53 (0,8)	0,27 (0,19)	126,8	0	***
<i>coord_to_comb</i>	0,27 (0,23)	0,2 (0,1)	105,6	0	***
<i>subord_to_comb</i>	0,72 (0,24)	0,8 (0,1)	117,5	0	***
<i>coord_to_sent</i>	0,28 (0,25)	0,36 (0,22)	73,5	0	***
<i>subord_to_sent</i>	0,77 (0,48)	1,43 (0,51)	1095,4	0	***
<i>avg_np_len</i>	3 (0,52)	2 (0,28)	2194,8	0	***
<i>comp_np_ratio</i>	0,66 (0,09)	0,53 (0,08)	1617,7	0	***

It is worth noting that graph descriptions are characterized by longer clauses, T-units, and sentences. On the other hand, essays exhibit a higher degree of embeddedness, as illustrated by the number of clauses per sentence (*cps*) and per T-unit (*cpt*). Argumentative essays tend to contain almost 1.5 subordinate clauses per sentence (*subord_to_sent*), while graph descriptions contain less than 1 (0.77, to be precise). One might speculate that this is due to the fact that argumentation is often constructed through the use of subordinating conjunctions, such as *although*, *because*, *since*, and so on.

Overall, our results support those of Panteleeva (2019: 38), who analyzed 23 syntactic complexity criteria and found that only average sentence depth and the Levenshtein distance between lemmatized sentences (neighboring and overall) were not statistically significant.

6.2. L2: Influence of Proficiency Level

Different corpora use different proficiency scales; for example, ACTR utilizes the Common European Framework of Reference for Languages (CEFR) with levels ranging from A1 (beginner) to C2 (superior) (Council of Europe, 2001). RULEC, on the other hand, relies on the standard of the American Council on the Teaching of Foreign Languages (ACTFL, 2012), which includes the levels Novice, Intermediate, Advanced, and Superior. Each level is further divided into Low, Mid, and High. As for RLC, it contains a mixture of the two systems.

Fortunately, ACTFL levels can be converted into the CEFR system according to ACTFL (2016). Tschirner et al. (2012) demonstrated the reliability of such a conversion, albeit for Oral Proficiency Interviews. We use the conversion described in ACTFL (2016) for productive skills. Table 3 shows which CEFR levels correspond to each of the ACTFL ones.

Table 3. Correspondences between ACTFL and CEFR proficiency scales (ACTFL, 2016: 4).

ACTFL	CEFR
Novice Low (NL)	0
Novice Mid (NM)	0 ³⁰
Novice High (NH)	A1
Intermediate Low (IL)	A2
Intermediate Mid (IM)	B1.1
Intermediate High (IH)	B1.2
Advanced Low (AL)	B2.1
Advanced Mid (AM)	B2.2
Advanced High (AH)	C1
Superior	C2

In practice, fine-grained CEFR levels (such as B1.1) are not normally used, which makes conversion back to ACTFL levels impossible.

The only learner English corpus in our dataset, REALEC, uses grades instead of proficiency levels. As the conversion from grades to levels is far from straightforward, we analyze L2 English and L2 Russian texts separately (refer to Section 7 for a description of the relationship between syntactic measures and grades). Unfortunately, this discrepancy in proficiency scales makes direct comparison of non-native Russian and English texts impossible.

Table 4 displays the number of texts written by Russian learners of different proficiency (only L1 texts and L2 texts with a specified proficiency level are included). There was only one Novice Low text, which we excluded from our analysis.

Table 4. Number and percentage of Russian texts by proficiency level.

Proficiency level	Number of texts (%)
A0	18 (0.08%)
A1	111 (0.46%)
A2	252 (1.05%)
B1	1,449 (6.06%)

³⁰ We mark NM as A0 for simplicity.

B2	1,528 (6.39%)
C1	49 (0.21%)
C2	13 (0.05%)
Native	20,475 (85.69%)
Total	23,895 (100%)

We present the mean complexity values by proficiency level and the significance of the results in Table 5. All the measures are highly significant ($p < 0.001$).

Table 5. Complexity measures by proficiency level (for Russian) and ANOVA results.

Complexity Measure	Mean (std)								F-statistic	p-val., sign. level
	A0	A1	A2	B1	B2	C1	C2	Native		
<i>msl</i>	10 (5)	9 (6)	11 (5)	14 (7)	15 (5)	14 (4)	13 (2)	15 (9)	41,1	0 ***
<i>mcl</i>	5 (1,01)	5 (1,2)	5 (1,16)	5 (1,2)	6 (1,28)	6 (1,31)	5 (0,81)	5 (2)	37,7	0 ***
<i>mtl</i>	6 (3)	6 (2)	7 (2)	9 (2)	9 (2)	9 (2)	7 (1,5)	8 (3)	72,2	0 ***
<i>cps</i>	2 (0,76)	1,94 (1,37)	2 (0,95)	3 (1,12)	3 (0,83)	2 (0,51)	3 (0,64)	3 (1,72)	18,1	0 ***
<i>cpt</i>	1,33 (0,26)	1,3 (0,4)	1,41 (0,31)	1,64 (0,34)	1,63 (0,32)	1,65 (0,23)	1,49 (0,17)	1,45 (0,24)	35,8	0 ***
<i>lev_pos</i>	10 (9)	9 (11)	10 (5)	13 (7)	13 (5)	13 (4)	12 (2)	15 (12)	22,2	0 ***
<i>lev_dep</i>	11 (9)	9 (11)	10 (6)	14 (7)	15 (5)	14 (4)	13 (3)	16 (13)	29,3	0 ***
<i>mtd</i>	4 (0,79)	4 (0,86)	4 (0,8)	5 (1,01)	5 (0,88)	5 (1,05)	5 (0,6)	5 (0,98)	114,5	0 ***
<i>mdtd</i>	4 (0,79)	4 (0,9)	4 (0,85)	5 (1,07)	5 (0,98)	5 (1,06)	5 (0,65)	5 (1,08)	100,8	0 ***
<i>mxtd</i>	6 (1,93)	5 (1,71)	6 (1,64)	8 (2)	8 (1,7)	8 (2)	8 (1,05)	9 (2)	74,1	0 ***
<i>mntd</i>	2 (0,69)	3 (0,72)	2 (0,7)	3 (0,98)	3 (1,03)	3 (1,1)	3 (0,63)	2 (0,85)	26,1	0 ***
<i>mdd</i>	2 (0,53)	2 (0,67)	2 (0,45)	3 (0,47)	3 (0,38)	3 (0,35)	3 (0,27)	3 (3)	19,2	0 ***
<i>node_to_term</i>	0,97 (0,11)	1,02 (0,16)	0,95 (0,13)	0,97 (0,13)	0,99 (0,14)	0,98 (0,14)	0,88 (0,14)	0,98 (0,13)	8,5	0 ***
<i>coord_to_subord</i>	1,14 (0,62)	1,36 (1,51)	1,16 (1,59)	0,78 (0,79)	0,82 (0,95)	0,59 (0,45)	1,08 (0,72)	1,44 (1,67)	10,6	0 ***

<i>coord_to_comb</i>	0,44 (0,21)	0,52 (0,33)	0,45 (0,24)	0,38 (0,17)	0,39 (0,17)	0,33 (0,15)	0,48 (0,14)	0,53 (0,16)	15,8	0 ***
<i>subord_to_comb</i>	0,45 (0,21)	0,42 (0,32)	0,54 (0,25)	0,61 (0,17)	0,61 (0,17)	0,67 (0,15)	0,52 (0,14)	0,47 (0,16)	27,8	0 ***
<i>coord_to_sent</i>	0,51 (0,43)	0,49 (0,95)	0,55 (0,53)	0,62 (0,48)	0,65 (0,46)	0,51 (0,31)	0,85 (0,4)	1,03 (1,2)	3,9	0,000 7 ***
<i>subord_to_sent</i>	0,53 (0,43)	0,45 (0,6)	0,65 (0,61)	1,04 (0,76)	1,01 (0,55)	0,96 (0,33)	0,9 (0,35)	0,88 (0,68)	27,4	0 ***
<i>avg_np_len</i>	1,68 (0,22)	1,66 (0,23)	1,65 (0,23)	1,92 (0,31)	2 (0,36)	1,99 (0,36)	1,75 (0,19)	1,99 (0,36)	108,2	0 ***
<i>comp_np_ratio</i>	0,46 (0,08)	0,48 (0,12)	0,44 (0,12)	0,53 (0,11)	0,58 (0,11)	0,56 (0,13)	0,49 (0,1)	0,54 (0,1)	74,4	0 ***

Interestingly, many values seem to display a curve in their development with a peak around the B2 level; this is true, for instance, for mean sentence length (*msl*) and mean clause length (*mcl*). Some values consistently increase through levels A0–B2 and suddenly drop at C1–C2 (e.g. mean T-unit length). On the one hand, this may be the case described by Housen et al. (2019: 9), who noted that “the growth in one domain [...] is likely to be accompanied by a plateau or decrease in another”. However, we believe a different explanation is more likely in this case. As evident from Table 4, the lowest and the highest proficiency levels are the two smallest categories in our dataset, with most texts pertaining to intermediate levels (B1 and B2). This is a common problem with learner corpora, also noted by Panteleva (2019) for REALEC. If we ignore the A0 and C2 levels on the basis of insufficient data, the developmental tendencies become clearer: *msl*, *mcl*, and *mtl* all increase with growing proficiency, as do the number of clauses per T-unit, tree depth, MDD, average NP length, and the proportion of complex nominals.

It is noteworthy that most measures converge with native values at around B2–C1 proficiency (which is consistent with Kisselev et al.’s (2022) findings). However, mean sentence- and T-unit length, as well as Levenshtein distance for both POS and *deprel*, are higher for native speakers than for learners of any proficiency. This means that native speakers produce longer and more structurally diverse utterances.

The ratio of coordinate to subordinate clauses starts off at around 1.2 and slowly decreases to 0.58 at the C1 level; however, native Russian speakers seem to use coordination and subordination equally (the mean ratio is exactly 1). Comparing the *coord_to_sent* and *subord_to_sent* ratios, one will notice that learners of all proficiency levels use coordination roughly at the same rate (in every other sentence). The amount of subordination, however,

noticeably increases from around 0.5 (a subordinate clause for every other sentence) to around 1 (one subordinate clause per sentence). As noted by many previous researchers (Norris & Ortega, 2009; Lu, 2011; Kisselev et al., 2021), L2 development is often characterized by a shift from coordination to subordination and, finally, to nominalization. Kisselev et al. (2021) use nesting depth, similar to tree depth considered in our research, to gauge complexity at the phrasal level; along with growing NP length, the increase in mean and median tree depth does point to an increased reliance on nominalization by learners at higher proficiency levels.

6.3. L1: Influence of Language and Genre

To compare syntactic complexity measures of Russian and English texts, we used four native corpora: CoRST for formal Russian, LOCNESS for formal English, Pikabu for informal Russian, and Reddit for informal English. We ran four sets of ANOVAs, controlling for genre and language. The results are reported below.

6.3.1. Formal vs. Informal

Table 6 presents the mean syntactic complexity values for formal and informal native Russian texts. Overall, all measures are highly significant, except for *cps*, which has a p-value of 0.04, and MDD, which is not significant at all.

Table 6. Mean complexity values for formal and informal Russian texts and ANOVA results.

Complexity Measure	Mean (std)		F-statistic	p-value	Significance Level
	Formal Russian	Informal Russian			
<i>msl</i>	17 (7)	15 (9)	161,6	0	***
<i>mcl</i>	6 (4)	5 (1,9)	293,1	0	***
<i>mtl</i>	10 (5)	7 (3)	1061,9	0	***
<i>cps</i>	3 (0,99)	3 (1,77)	4,1	0,04349	*
<i>cpt</i>	1,63 (0,27)	1,43 (0,23)	1277,6	0	***
<i>lev_pos</i>	16 (7)	15 (13)	15,9	0,00007	***
<i>lev_dep</i>	18 (8)	16 (13)	41,8	0	***
<i>mtd</i>	6 (0,95)	5 (0,93)	2047,6	0	***
<i>mdtd</i>	6 (1,07)	5 (1,03)	1832	0	***

<i>mxt</i>	10 (2)	9 (2)	1075,5	0	***
<i>mnt</i>	3 (1,09)	2 (0,82)	338,6	0	***
<i>mdd</i>	3 (6)	3 (3)	0,6	0,42102	
<i>node_to_term</i>	1,07 (0,12)	0,97 (0,13)	1235	0	***
<i>coord_to_subord</i>	0,94 (1,36)	1,49 (1,69)	188,8	0	***
<i>coord_to_comb</i>	0,42 (0,15)	0,54 (0,16)	923,5	0	***
<i>subord_to_comb</i>	0,58 (0,15)	0,46 (0,16)	915	0	***
<i>coord_to_sent</i>	0,85 (0,57)	1,05 (1,25)	49,7	0	***
<i>subord_to_sent</i>	1,14 (0,59)	0,85 (0,68)	316,5	0	***
<i>avg_np_len</i>	2 (0,35)	1,96 (0,35)	1347,2	0	***
<i>comp_np_ratio</i>	0,61 (0,09)	0,53 (0,1)	980,8	0	***

There is a clear tendency of using longer sentences in formal writing. In addition, the clauses in formal language are more structurally diverse, judging from the *lev_pos* and *lev_dep* features. At the phrasal level, academic texts display a higher degree of embeddedness; the average NP length and the proportion of complex NPs are higher than in informal speech. Along with the higher proportions of subordinate clauses and lower ratios of coordinate clauses, this suggests that formal texts are more complex than informal ones. This is also consistent with Halliday's (1996) continuum of dynamic and synoptic styles, the former being represented by informal and the latter by formal texts.

For English, the results are reported in Table 7.

Table 7. Mean complexity values for formal and informal English texts and results of ANOVA.

Complexity Measure	Mean (std)		F-statistic	p-value	Significance Level
	Formal English	Informal English			
<i>msl</i>	21 (5)	19 (7)	19,5	0,00001	***
<i>mcl</i>	7 (1,07)	6 (2)	101,5	0	***
<i>mtl</i>	15 (3)	11 (3)	423,8	0	***
<i>cps</i>	3 (0,61)	3 (1,18)	30,2	0	***
<i>cpt</i>	2 (0,3)	1,91 (0,37)	89	0	***
<i>lev_pos</i>	19 (4)	19 (9)	0,2	0,68061	

<i>lev_dep</i>	21 (5)	21 (9)	0,5	0,47437	
<i>mtd</i>	5 (0,57)	5 (0,76)	115,7	0	***
<i>mdtd</i>	5 (0,63)	5 (0,86)	68,5	0	***
<i>mxttd</i>	9 (1,48)	8 (1,56)	142,2	0	***
<i>mntd</i>	3 (0,79)	3 (0,96)	10,8	0,00102	**
<i>mdd</i>	3 (0,25)	3 (0,93)	1,2	0,26504	
<i>node_to_term</i>	0,68 (0,05)	0,64 (0,07)	97,1	0	***
<i>coord_to_subord</i>	0,29 (0,14)	0,52 (0,4)	122,1	0	***
<i>coord_to_comb</i>	0,22 (0,08)	0,32 (0,12)	272,8	0	***
<i>subord_to_comb</i>	0,78 (0,08)	0,68 (0,12)	272,1	0	***
<i>coord_to_sent</i>	0,44 (0,23)	0,77 (0,57)	117	0	***
<i>subord_to_sent</i>	1,58 (0,47)	1,6 (0,77)	0,2	0,65852	
<i>avg_np_len</i>	2 (0,31)	1,79 (0,22)	1640,6	0	***
<i>comp_np_ratio</i>	0,54 (0,07)	0,41 (0,07)	1109,8	0	***

With the English language, the situation is a little different: neither type of Levenshtein distance is significant for distinguishing between formal and informal texts. Since this measure reflects how diverse the clauses are structurally, the absence of a difference is perhaps expected: English has a much more rigid word order than Russian does, and the range of possible structures is essentially the same regardless of genre. Similarly to Russian, the MDD metric does not show any differences between formal and informal texts. Otherwise, the tendencies related to clause and sentence length and to coordination/subordination are the same as for Russian.

6.3.2. English vs. Russian

We now turn to comparing texts of the same register, but in different languages.

Table 8. Mean complexity values for formal Russian and English texts and results of ANOVA.

Complexity Measure	Mean (std)		F-statistic	p-value	Significance Level
	Formal Russian	Formal English			
<i>msl</i>	21 (5)	17 (7)	94,3	0	***

<i>mcl</i>	7 (1,07)	6 (4)	18,4	0,00002	***
<i>mtl</i>	15 (3)	10 (5)	377,7	0	***
<i>cps</i>	3 (0,61)	3 (0,99)	0,3	0,55438	
<i>cpt</i>	2 (0,3)	1,63 (0,27)	865,7	0	***
<i>lev_pos</i>	19 (4)	16 (7)	46,9	0	***
<i>lev_dep</i>	21 (5)	18 (8)	46,4	0	***
<i>mtd</i>	5 (0,57)	6 (0,95)	52,8	0	***
<i>mdtd</i>	5 (0,63)	6 (1,07)	39,6	0	***
<i>mxttd</i>	9 (1,48)	10 (2)	89,8	0	***
<i>mntd</i>	3 (0,79)	3 (1,09)	0	0,87827	
<i>mdd</i>	3 (0,25)	3 (6)	0	0,82499	
<i>node_to_term</i>	0,68 (0,05)	1,07 (0,12)	3399,2	0	***
<i>coord_to_subord</i>	0,29 (0,14)	0,94 (1,36)	79,6	0	***
<i>coord_to_comb</i>	0,22 (0,08)	0,42 (0,15)	632,3	0	***
<i>subord_to_comb</i>	0,78 (0,08)	0,58 (0,15)	634,4	0	***
<i>coord_to_sent</i>	0,44 (0,23)	0,85 (0,57)	171,1	0	***
<i>subord_to_sent</i>	1,58 (0,47)	1,14 (0,59)	169,7	0	***
<i>avg_np_len</i>	2 (0,31)	2 (0,35)	0,4	0,54599	
<i>comp_np_ratio</i>	0,54 (0,07)	0,61 (0,09)	176,8	0	***

First of all, the values of length-based measures are higher for Russian in terms of words (*msl*, *mtl*, *mcl*), but not clauses per sentence (*cps* is not significant). The larger Levenshtein distances in Russian may stem from the fact that the clauses are longer; however, the less rigid word order may also contribute to this fact. Conclusions cannot be drawn without further testing. A way to verify whether clause length has a substantial effect on the average Levenshtein distance is to adjust the distance by the number of words in a clause. This is an interesting issue for future research.

MDD, minimum tree depth, and average NP length did not reach significance. Mean, median, and maximum tree depth, however, show a higher degree of nesting in formal English texts than in Russian ones. Interestingly, speakers of Russian seem to rely more heavily on subordination than coordination, while the opposite is true for English speakers.

Table 9. Mean complexity values for informal Russian and English texts and results of ANOVA.

Complexity Measure	Mean (std)		F-statistic	p-value	Significance Level
	Informal Russian	Informal English			
<i>msl</i>	19 (7)	15 (9)	3145,7	0	***
<i>mcl</i>	6 (2)	5 (1,91)	1234,1	0	***
<i>mtl</i>	11 (3)	7 (3)	14693,1	0	***
<i>cps</i>	3 (1,18)	3 (1,77)	887,7	0	***
<i>cpt</i>	1,91 (0,37)	1,43 (0,23)	23278	0	***
<i>lev_pos</i>	19 (9)	15 (13)	1266,3	0	***
<i>lev_dep</i>	21 (9)	16 (13)	1546,7	0	***
<i>mtd</i>	5 (0,76)	5 (0,93)	493,9	0	***
<i>mdtd</i>	5 (0,86)	5 (1,03)	974,4	0	***
<i>mxtd</i>	8 (1,56)	9 (2)	703,5	0	***
<i>mntd</i>	3 (0,96)	2 (0,82)	533,8	0	***
<i>mdd</i>	3 (0,93)	3 (3)	5,6	0,01821	*
<i>node_to_term</i>	0,64 (0,07)	0,97 (0,13)	90670	0	***
<i>coord_to_subord</i>	0,52 (0,4)	1,49 (1,69)	5822,1	0	***
<i>coord_to_comb</i>	0,32 (0,12)	0,54 (0,16)	22642,9	0	***
<i>subord_to_comb</i>	0,68 (0,12)	0,46 (0,16)	22608,2	0	***
<i>coord_to_sent</i>	0,77 (0,57)	1,05 (1,25)	768,4	0	***
<i>subord_to_sent</i>	1,6 (0,77)	0,85 (0,68)	9781,6	0	***
<i>avg_np_len</i>	1,79 (0,22)	1,96 (0,35)	3179,3	0	***
<i>comp_np_ratio</i>	0,41 (0,07)	0,53 (0,1)	16421	0	***

The tendencies described for the formal register are upheld by the values calculated on the basis of informal texts. Phrasal complexity measures (*node_to_term*, *avg_np_len*, *comp_np_ratio*) are higher for English than for Russian, – a trend that was also true but not as noticeable in the formal register.

The discrepancy between coordination and subordination, on the other hand, is even more pronounced here (the *coord_to_subord* ratio is 0.5 for Russian and 1.5 for English). The

coord_to_sent and *subord_to_sent* measures show diametrically opposed trends for Russian and English (repeating what was described for formal texts above).

6.4. Are Heritage and Foreign Texts different?

To investigate the differences between heritage and foreign writing, we used data from three Russian corpora: ACTR, RLC, and RULEC. In total, 2335 texts were included in the analysis (567 heritage, 1768 native). Table 10 presents the results.

Table 10. Mean complexity values for HL and FL speakers and results of ANOVA.

Complexity Measure	Mean (std)		F-statistic	p-value	Significance Level
	HL	FL			
<i>msl</i>	13 (7)	14 (4)	11,4	0,00075	***
<i>mcl</i>	5 (1,21)	5 (1,08)	79,8	0	***
<i>mtl</i>	8 (2)	9 (2)	60,3	0	***
<i>cps</i>	3 (1,15)	3 (0,84)	0,2	0,67103	
<i>cpt</i>	1,56 (0,35)	1,58 (0,27)	1,1	0,28649	
<i>lev_pos</i>	12 (7)	12 (5)	3,2	0,07478	
<i>lev_dep</i>	13 (8)	14 (5)	6,5	0,01088	*
<i>mtd</i>	5 (1,07)	5 (0,87)	45,6	0	***
<i>mdtd</i>	4 (1,1)	5 (0,97)	54,3	0	***
<i>mxtd</i>	7 (2)	8 (1,62)	3,8	0,05084	
<i>mntd</i>	3 (0,85)	3 (0,9)	46,6	0	***
<i>mdd</i>	3 (0,5)	3 (0,34)	1,4	0,23012	
<i>node_to_term</i>	0,94 (0,13)	0,97 (0,15)	18,1	0,00002	***
<i>coord_to_subord</i>	0,91 (1,01)	0,83 (1,05)	2,9	0,08975	
<i>coord_to_comb</i>	0,42 (0,2)	0,39 (0,18)	7,3	0,00677	**
<i>subord_to_comb</i>	0,58 (0,2)	0,61 (0,19)	9,5	0,00213	**
<i>coord_to_sent</i>	0,63 (0,55)	0,62 (0,5)	0,1	0,72015	
<i>subord_to_sent</i>	0,93 (0,76)	0,92 (0,48)	0,1	0,7022	
<i>avg_np_len</i>	1,76 (0,31)	1,95 (0,33)	161	0	***
<i>comp_np_ratio</i>	0,47 (0,12)	0,54 (0,11)	152,1	0	***

Many complexity values turned out to be insignificant for distinguishing between HL and FL speakers. Among those are the number of clauses per sentence and per T-unit, the ratio of coordinate and subordinate clauses per sentence, and the ratio of coordinate to subordinate clauses. POS-based Levenshtein distance was also insignificant, while its *deprel*-based counterpart had a p-value of 0.01.

Overall, the measures of coordination and subordination turned out to be less relevant than could have been hypothesized on the basis of previous research; for instance, Kisselev et al. (2022) found a discrepancy in the developmental tendencies related to coordinate clauses. However, the inconsistencies between Kisselev et al.'s (2022) findings and ours can be explained by the fact that the authors looked at different proficiency levels, while we did not.

In general, FL speakers use longer clauses and sentences and display a higher degree of embeddedness, as illustrated by mean and median tree depth and the Node-to-Terminal-Node ratio. All in all, several measures signal that FL speech is somewhat more complex, although this cannot be said with confidence until different proficiency groups are analyzed.

7. Predicting Proficiency

As a final step, we turn to the task of predicting proficiency scores on the basis of syntactic complexity. Given the two types of proficiency indices (CEFR levels vs. grades) present in our dataset, we pose and attempt to solve both a classification and a regression problem.

7.1. Classification

The task of determining a learner’s proficiency can be approached as a multi-class classification problem, the target classes being represented by the categorical proficiency values listed in Table 3. Once again, only the Russian part of the dataset contains proficiency levels.

A closer look at the data revealed a number of problems. Firstly, some complexity values could not be calculated for the entire dataset: for example, ratio-based measures such as *coord_to_subord*, *coord_to_comb*, *subord_to_comb* require the denominator to be greater than zero. We replace these missing values with 0.

The second problem is the pronounced class disbalance: the *native* class takes up over 85% of all proficiency-tagged texts. We deal with this by applying oversampling techniques to the training set via the *imblearn* package for Python³¹. All syntactic complexity measures are used as features. We perform a train/test split stratified by proficiency level with 25% of the data allocated to the test set.

7.1.1. Logistic Regression

By training a simple Logistic Regression model implemented in the *scikit-learn* Python library³² (with *class_weight*=“*balanced*” and other parameters left at default), we achieve a weighted F1-score of 0.57. The classification report is displayed in Figure 8, and Figure 9 presents the confusion matrix.

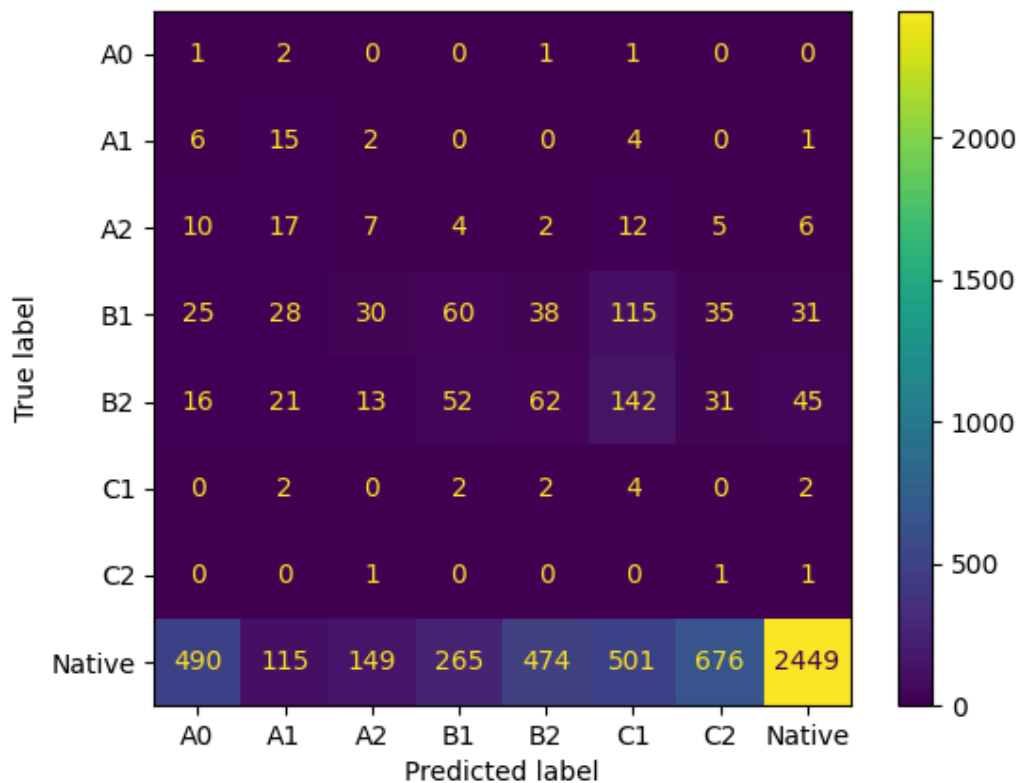
³¹ <https://pypi.org/project/imblearn/>

³² <https://pypi.org/project/scikit-learn/>

Figure 8. Classification report for a simple Logistic Regression model.

	precision	recall	f1-score	support
A0	0.00	0.20	0.00	5
A1	0.07	0.54	0.13	28
A2	0.03	0.11	0.05	63
B1	0.16	0.17	0.16	362
B2	0.11	0.16	0.13	382
C1	0.01	0.33	0.01	12
C2	0.00	0.33	0.00	3
Native	0.97	0.48	0.64	5119
accuracy			0.44	5974
macro avg	0.17	0.29	0.14	5974
weighted avg	0.84	0.44	0.57	5974

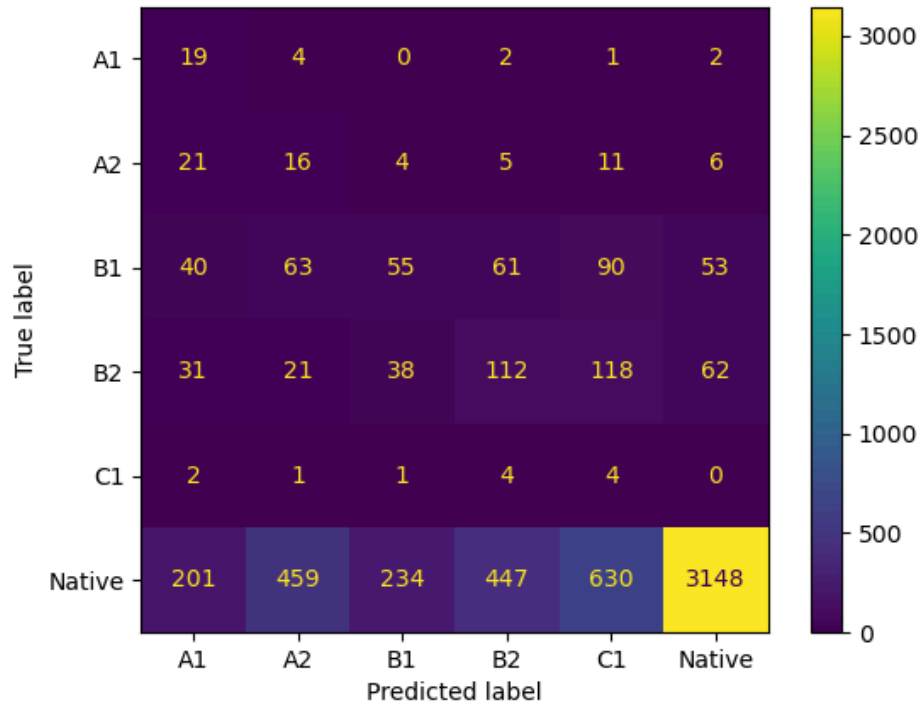
Figure 9. Confusion matrix of proficiency labels.



It is clear that the smallest classes are barely recognized at all, and even the biggest classes (Native, B1 and B2) are often confused with others. The *native* class is most often confused with high-proficiency learner texts, C1 and C2; however, an unexpectedly high number of native texts is attributed to the lowest level (A0). We hypothesize that this is caused by the extremely low number of A0 texts in the corpus. Deleting the two smallest classes (A0 and C2) increases weighted F1-score by 10%, to 0.67; however, the score on

smaller classes increases only slightly to 0.15-0.20, with the *native* class contributing most of the correctly identified instances. Figure 10 displays the confusion matrix for the improved model.

Figure 10. Confusion matrix of proficiency labels (improved Logistic Regression model).



As evidenced by Figure 10, the new model is more successful at predicting the levels A1, B1, B2, and Native, although adjacent levels are still often confused. A strikingly high number of Native essays is attributed to A2; there may be some underlying similarities between the two not shown by ANOVAs, or the model simply inferred wrong correlations due to the small size of the A2 class.

Running a grid search algorithm did not improve the results, proving that the parameters were already optimal. As an experiment, we added a new set of features: the proportions of all the different types of clauses in the text (these types were described in Section 4.1). However, the F1-score decreased back to 0.56. This suggests that using fine-grained complexity measures, as suggested by Robinson and Ellis (2008), is not always beneficial, at least not in the task of automatic proficiency determination.

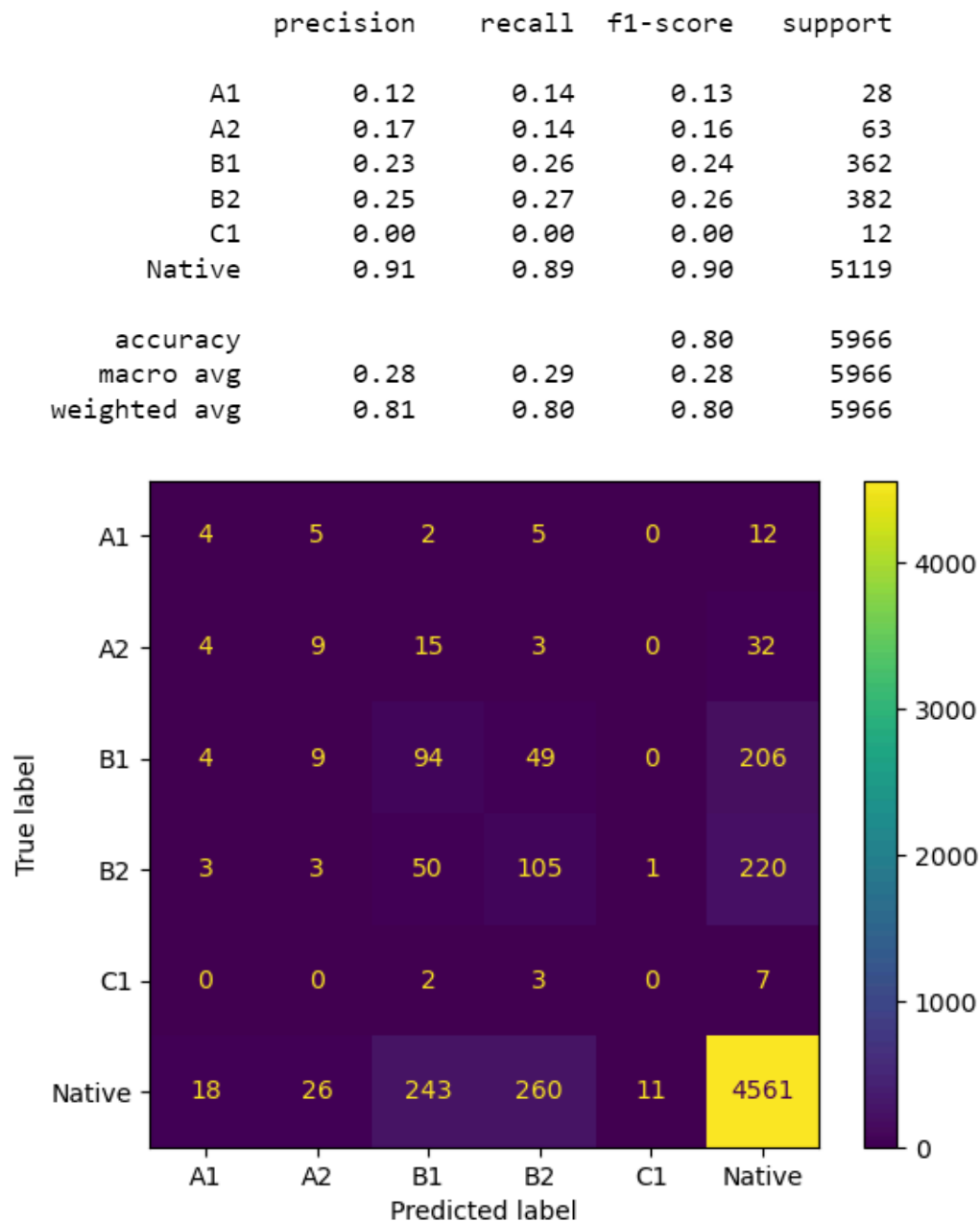
7.1.2. Decision Tree Classifier

Decision trees can be more efficient when the relationship between features and the target variable is complex and/or non-linear. Predicting proficiency on the basis of text complexity seems like the perfect use case for this type of algorithm, since many researchers

noted the intricacy of the relationship between complexity and proficiency (Wolfe-Quintero et al., 1998; Ortega, 2003).

Training a *scikit-learn* Decision Tree Classifier with balanced class weights yielded a weighted F1-score of 0.8. Figure 11 shows the classification report and the confusion matrix of the new model.

Figure 11. Classification report and confusion matrix of the Decision Tree Classifier.



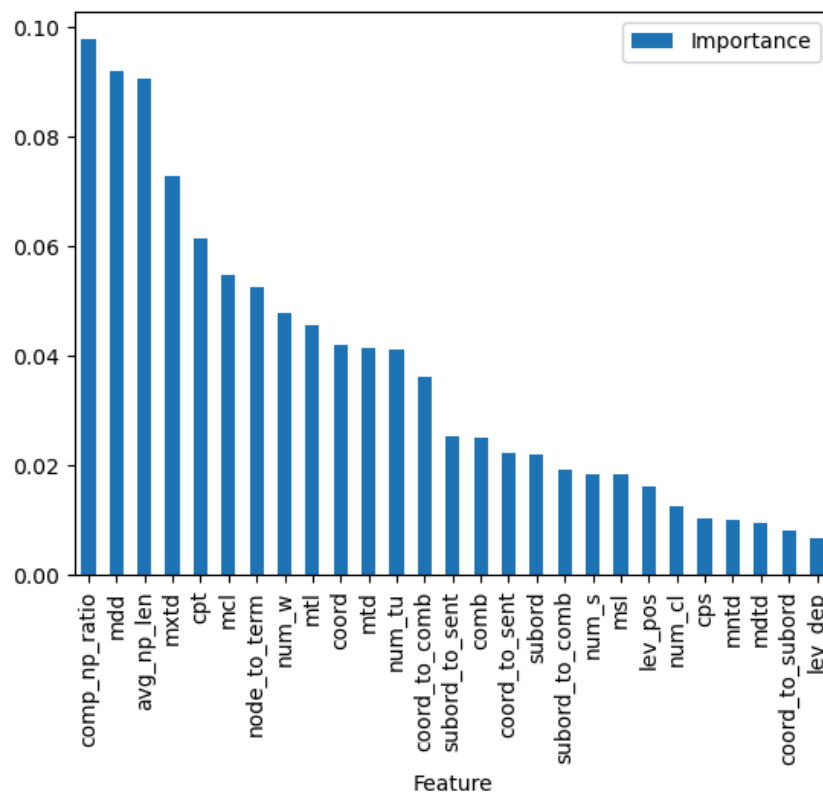
Clearly, for smaller classes the quality remains unsatisfactory. Panteleeva (2019), working with the data of a different corpus (REALEC), faced the same problem when trying to predict grades based on text complexity (she treated grades as a categorical rather than

continuous variable). To improve the results, she tried combining the essays into two categories, *best* and *non-best*, and treating the task as a binary classification problem. By also adding text vectors obtained from TF-IDF Vectorizer, she managed to achieve an F1-score of 0.85; however, the smaller *best* class was barely recognized at all (Panteleeva, 2019: 47–50).

7.1.3. Feature Importance

Despite the mediocre quality of both classification models, it is interesting to see which features were deemed important by the algorithm (we only analyze Decision Tree Classifier as the more successful one). By default, feature importance for tree-based models is measured on the basis of Mean Decrease in Impurity (MDI), also called Gini Importance. It reflects the number of times a feature gets used for splitting a node, weighted by the number of samples split by it (Raileanu & Stoffel, 2004). These coefficients are plotted in Figure 12.

Figure 12. Importance of various features for the Decision Tree Classifier.



Phrasal complexity measures, such as complex NP ratio and average NP length, were deemed the most important, along with Mean Dependency Distance. Mean clause and T-unit length are also quite significant, while mean sentence length is not.

It is noteworthy that coordination and subordination measures seem to be grouped together and are only moderately important. Interestingly, the least significant feature is the

deprel-based Levenshtein distance; recall that this measure was the least significant for distinguishing between task types (Section 6.1).

When using 12 most important features (out of 27 total) and re-training the Decision Tree Classifier, the F1-score decreases to 0.76; thus, removing more than half of the features only results in a 4% loss of model quality.

All in all, further work is clearly needed to refine the classification algorithm. The crucial steps are increasing the size of the smaller (non-native) classes and ascertaining the correctness of the proficiency labels (e.g., by re-checking a portion of the dataset and calculating annotator agreement between experts). If a sufficient amount of data is collected, more sophisticated models can be trained; one of the possible approaches is fine-tuning a BERT model for the task of classification. Using few-shot learning with state-of-the-art Large Language Models (LLMs) such as OpenAI GPT-4 is another promising direction of future research, as it does not require a large training set.

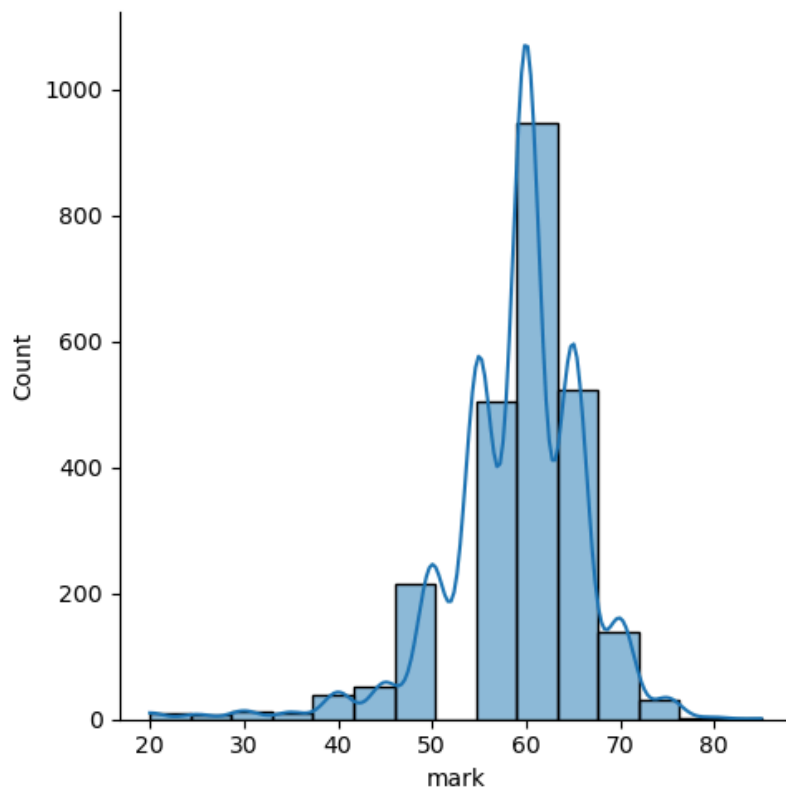
The feature correlations uncovered in Section 5 show that the syntactic complexity measures we use as features are not independent. With this in mind, experimenting with a Hierarchical Logistic Regression (HLR) model seems appropriate, since such algorithms are capable of capturing feature nestedness and class hierarchy (Wong & Mason, 1985).

7.2. Regression

For REALEC, which uses continuous grades instead of levels to denote proficiency, we implement a regression algorithm. The motivation to use regression instead of classification comes from Ortega (2012), who argues that both proficiency and complexity scores should be represented as interval scales.

Some grades are on a scale from 0 to 10, others on a scale from 0 to 100, with some being multiples of five. We convert all grades to a 0–100 scale; the distribution is illustrated in Figure 13.

Figure 13. Distribution of grades in REALEC.



As with proficiency levels, there are very few grades on either extreme: one text received an 85, three received an 80, 16 received either a 20 or a 25, while the overwhelming majority got a score from 55 to 65 points.

A *scikit-learn* Linear Regression model with default parameters yielded an R^2 -score of 0.15, which means that the variance in the data is poorly explained by the model. The mean absolute error (MAE) was 5, and the root mean square error (rMSE) was 7. Recall that the grades in our data are multiples of 5; on average, the grade predicted by the model is 5 points off, meaning that adjacent grades are often confused. This problem might be caused by the properties of our dataset more than by those of the model itself: Panteleeva (2019: 6) reports a 0.33 annotator agreement (measured by Cohen’s kappa) between expert grades of 50 REALEC essays.

Additional experiments involved Ridge, Lasso, and ElasticNet models, as well as a grid search for optimal parameters. The choice of Ridge regression, which uses L2 regularization as a penalty, is particularly useful when multicollinearity of features is present (which was demonstrated in Section 5). Lasso, on the other hand, is based on L1 regularization and is useful when some features in the dataset are more important than others (i.e., it automatically selects the most important features). Based on the analysis of the classification model features

in Section 7.1.3, we believed that the Lasso regression might also be beneficial for the task of predicting continuous grades. As for ElasticNet, it utilizes both types of penalty, combining the advantages of the other two models.

However, our experiments did not help improve the R^2 score or MAE/rMSE. This may be due either to inconsistencies in the assignment of grades to the essays or the fact that the variance in grades is explained by factors other than syntactic complexity.

Future experiments may involve adding text embeddings, although that will mean abandoning the idea of predicting grades solely on the basis of syntactic complexity parameters. Further improvements can be made by fine-tuning a pretrained BERT-based model, as was recently done by Bannò et al. (2022) and Zheng et al. (2022) for L2 English, albeit for oral data. Few-shot learning with LLMs is rarely used for regression, but the task of predicting grades can also be viewed as a multi-class classification problem, as was done by Panteleeva (2019).

8. Future Work and Limitations

In this research, we tried to offer a comprehensive perspective of the nature of syntactic complexity on the material of Russian and English. However, a lot is left to be explored. First of all, we did not take into account oral texts; while some progress has been made in understanding the specifics of oral complexity in L2 English (Larsen-Freeman, 2006; Ellis, 2009; Lu, 2012), to the best of our knowledge, no similar studies exist for Russian.

Another interesting direction of research is comparing child language acquisition to SLA in order to trace developmental trends and determine whether there are any similarities. Unsworth and Blom (2010) discuss a number of important conceptual and methodological problems related to comparing children learning their first language to children and adults learning an L2. In general, differences are to be expected, since L2 learners have a pre-existing language system, while L1 learners do not.

Several limitations are related to the datasets that were at our disposal during this study. First of all, the L2 corpora are small in size, complicating statistical analysis and application of machine learning techniques. Second, the variety of approaches to documenting speaker meta-information made a more thorough analysis of extralinguistic factors impossible. For instance, we were not able to test L1 influence, since full information on the speaker's L1 was only available in one corpus (namely REALEC). Next, a more fine-grained approach to speaker types might be warranted: for example, Ortega (2003) found that there were differences in the syntactic complexity values of texts produced by FL and SL learners. In the corpora we used, there was no clear indication of speaker type, so we were not able to investigate this matter. With that said, we believe that a more thorough approach to collecting and documenting extralinguistic information would be extremely beneficial.

Furthermore, even though our analysis included 27 measures of syntactic complexity, this list is far from exhaustive. More could be understood on the nature of phrasal complexity by employing constituency parsing or adding error analysis (for example, by examining which syntactic structures are challenging to learners and in which ways).

9. Conclusion

In this study, we explored a number of complexity measures pertaining to different levels of syntax, namely sentential, clausal, and phrasal. We analyzed the correlations between these features, confirming that the relationships hold for both L1 and L2 texts in Russian and English (RQ1). Length-based measures, such as mean length of clause or sentence, were found to be highly positively correlated. Two other measures, namely the ratios of coordinate and subordinate clauses to all combined clauses, displayed a strong negative correlation, providing support for Bulté and Housen's (2012) criticism of the Coordination Index. The numbers of coordinate and subordinate clauses per sentence, which they proposed as substitutes, had a much lower correlation coefficient.

Our findings in regard to the second research question (on the relationship of proficiency and complexity) are consistent with previously those reported in the literature (Ortega, 2003; Kisselev et al., 2022; Panteleeva, 2019). For example, we discovered that many complexity values (such as average sentence and clause length, the number of clauses per sentence, and the degree of phrasal embeddedness as illustrated by tree depth and the proportion of complex nominals) slowly increased with growing proficiency and plateaued around the Upper Intermediate–Advanced (B2–C1) level. Unfortunately, we were only able to analyze these relationships for the Russian language, since our only learner English corpora contained examination grades instead of proficiency values. Moreover, the lack of low- and high-proficiency texts might have obscured the full picture of the developmental trends.

In answering the RQ3 on the interaction of task type and complexity, we turned to the IELTS written examination essays written by Russian learners of English. Our results align with those reported by Panteleeva (2019). All 20 syntactic complexity measures were suitable for distinguishing between graph descriptions and argumentative essays, – a conclusion also supported by Ortega (2003) who reported systematic complexity differences with regard to genre and instructional setting.

A look at the features of texts differing by genre and language (RQs 4 and 5) revealed several important tendencies. First, formal texts consist of longer sentences with a higher degree of embeddedness. In academic Russian, the clauses are more structurally diverse than in informal writing, but the same does not hold for English. We hypothesize that this is due to the more rigid word order patterns of the English language. Both in formal and informal

writing, speakers of Russian rely more heavily on subordination, while English speakers prefer coordination.

Finally, we turn to the question of what syntactic complexity features are the most useful in predicting a learner’s proficiency (RQ6). Based on the data that were available, we implemented a classification algorithm for Russian L2 texts and a regression model for English ones. Both models suffered from problems inherent to the corpora we used; more specifically, there was a high class imbalance and very few texts of extremely low or high proficiency. As has been noted in the literature (e.g. Ortega, 2003), the relationship between complexity and proficiency is not always linear, which might have also presented a challenge for the models. Overall, for predicting a categorical proficiency value, measures of phrasal complexity and length-based indices were the most useful.

The novelty of our work lies in the comprehensive approach to syntactic complexity, spanning across different languages, proficiency levels, and genres. By performing an extensive analysis of 20 complexity measures, we demonstrate their robustness in distinguishing between different types of texts across both L1 and L2 texts in Russian and English. We also propose a new measure of structural diversity, namely average Levenshtein distance between sequences of part-of-speech and dependency relationship tags.

The practical value of our research is enhanced by the development of *syntaxcomp* – a Python tool capable of extracting a variety of language-independent syntactic complexity measures from CoNLL-U annotations. This tool not only facilitates the replication of our results but also empowers other researchers to conduct similar analyses for different languages or corpora.

Further research is needed to fully understand the intricate relationship of syntactic complexity with language, genre, speaker type, and proficiency. We hope that our findings will contribute to the development of the field, while the availability of *syntaxcomp* will attract more researchers from various areas of linguistics to the study of syntactic complexity.

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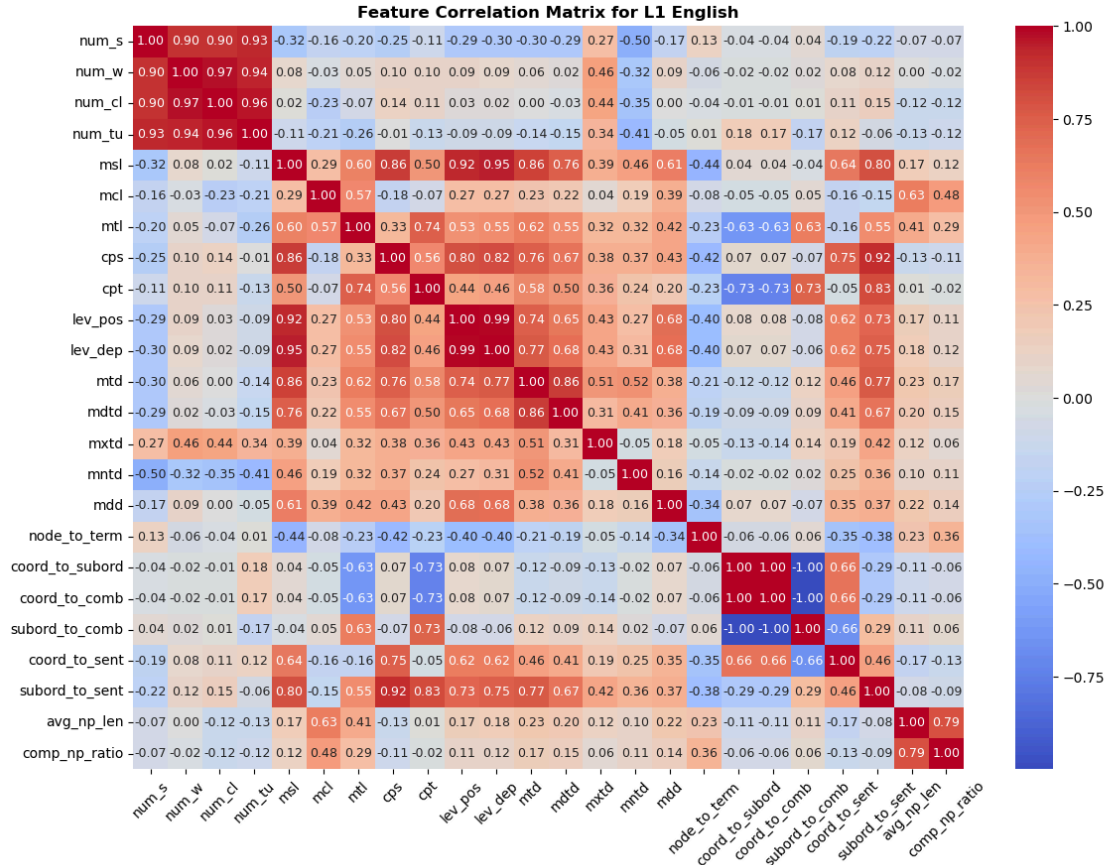
Appendices

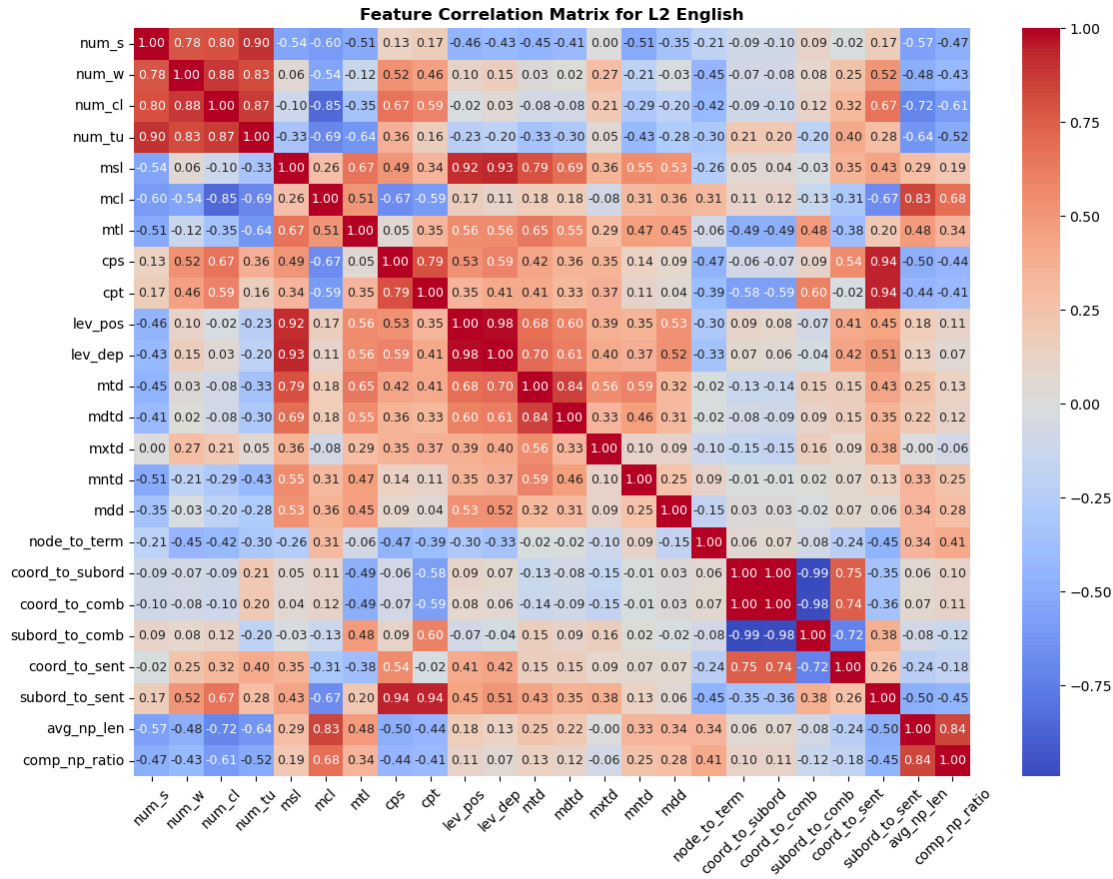
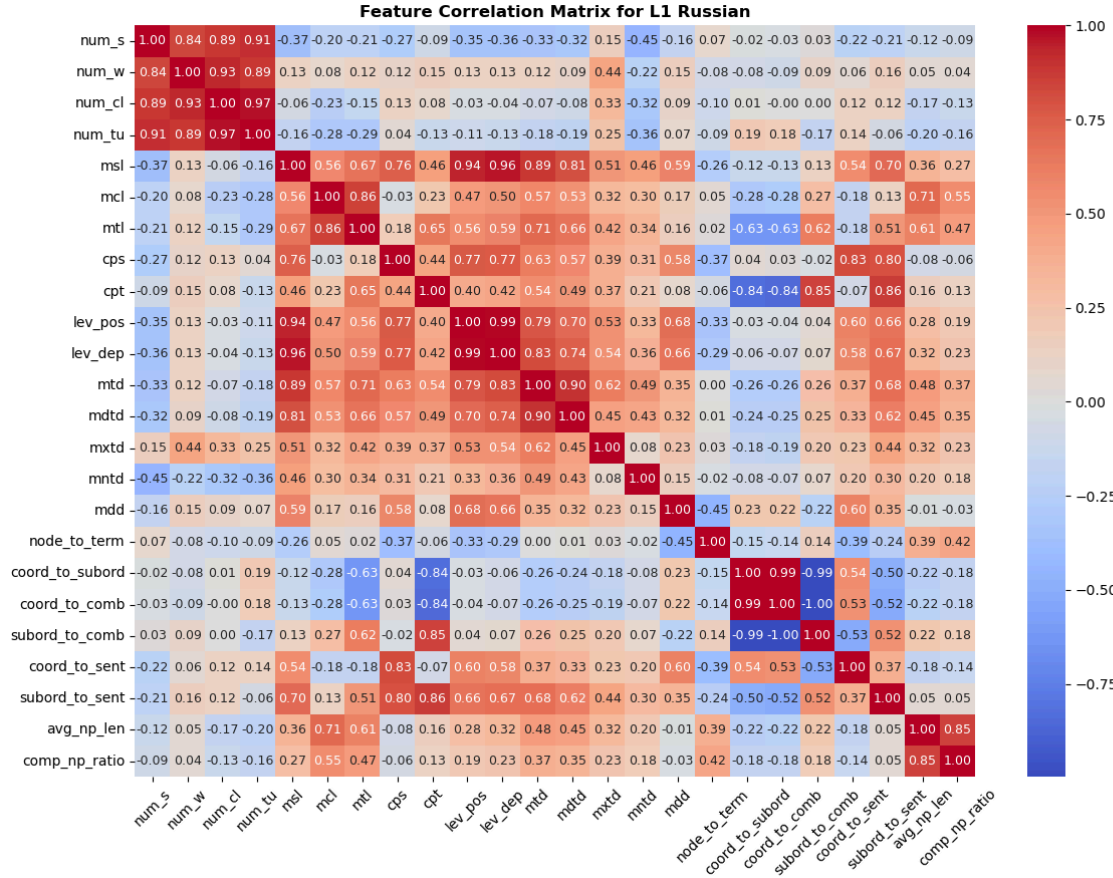
Appendix 1. List of complexity measures, their abbreviations, and formulas.

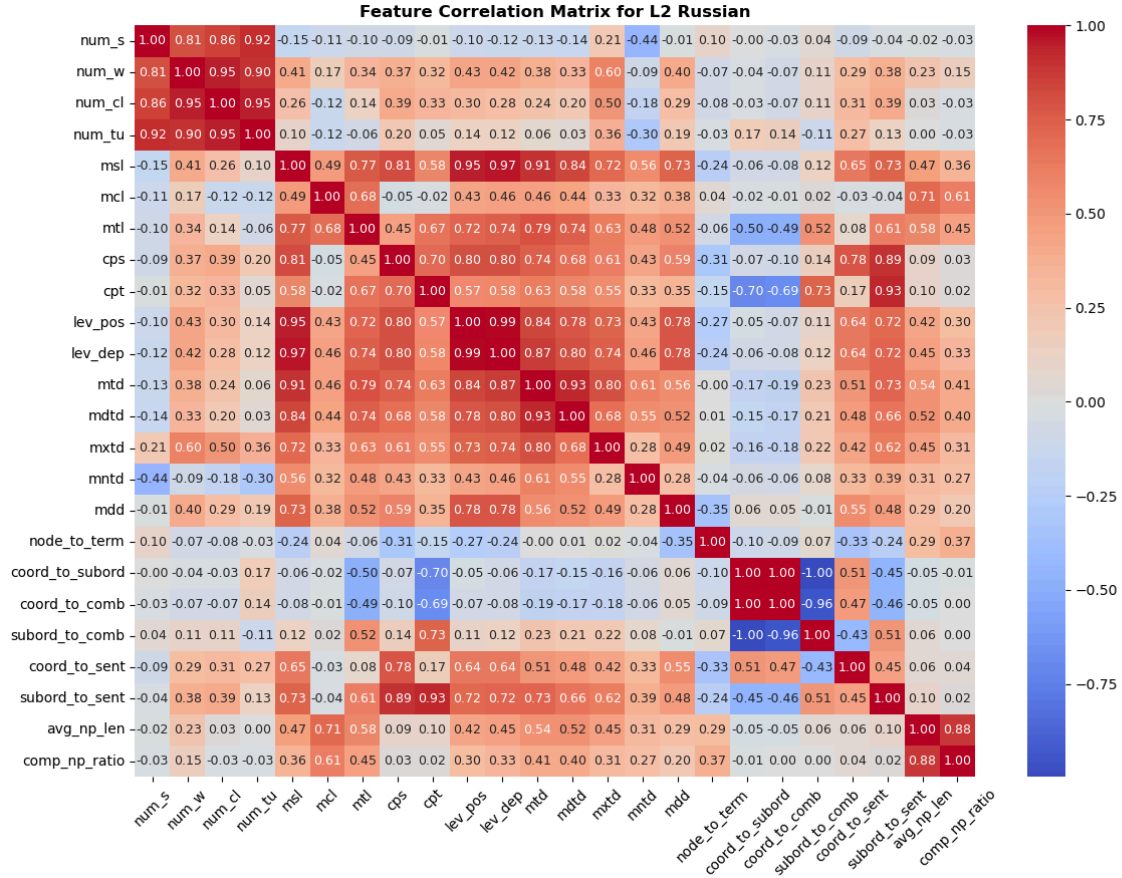
Type	Measure	Abbreviation	Formula
absolute length-based measures	number of sentences	<i>num_s</i>	–
	number of words	<i>num_w</i>	–
	number of clauses	<i>num_cl</i>	–
	number of T-units	<i>num_tu</i>	–
	number of combined clauses	<i>comb</i>	$num_cl - num_s$
	number of coordinate clauses	<i>coord</i>	clauses with <i>deprel</i> ‘conj’ or ‘parataxis’
	number of subordinate clauses	<i>subord</i>	$comb - coord$
relative length-based measures	mean sentence length (words)	<i>msl</i>	num_w / num_s
	mean clause length (words)	<i>mcl</i>	num_w / num_cl
	mean T-unit length (words)	<i>mtl</i>	num_w / num_tu
	average sentence length (clauses)	<i>cps</i>	num_cl / num_s
	average T-unit length (clauses)	<i>cpt</i>	num_cl / num_tu
clausal level	average pairwise Levenshtein distance (for POS)	<i>lev_pos</i>	<i>mean</i> (average distance between POS sequences for every pair of sentences)
	average pairwise Levenshtein distance (for <i>deprel</i>)	<i>lev_dep</i>	<i>mean</i> (distance between <i>deprel</i> sequences for every pair of sentences)
sentential level	mean tree depth	<i>mtd</i>	<i>mean</i> (distance from root to farthest leaf for each sentence)
	median tree depth	<i>mdtd</i>	<i>median</i> (distance from root to farthest leaf for each sentence)
	maximum tree depth	<i>mxttd</i>	<i>max</i> (distance from root to farthest leaf for each sentence)
	minimum tree depth	<i>mntd</i>	<i>min</i> (distance from root to farthest leaf for each sentence)

	mean dependency distance	<i>mdd</i>	<i>mean</i> (dependency distance between every token and its head)
	node to terminal node ratio	<i>node_to_term</i>	number of nonterminal nodes / number of terminal nodes
coordination / subordination	coordinate to subordinate clause ratio	<i>coord_to_subord</i>	<i>coord / subord</i>
	coordinate to combined clause ratio	<i>coord_to_comb</i>	<i>coord / comb</i>
	subordinate to combined clause ratio	<i>subord_to_comb</i>	$(comb - coord) / comb$
	coordinate clause to sentence ratio	<i>coord_to_sent</i>	<i>coord / num_s</i>
	subordinate clause to sentence ratio	<i>subord_to_sent</i>	$(comb - coord) / num_s$
phrasal level	complex NP ratio	<i>comp_np_ratio</i>	(number of NPs with length > 1) / number of NPs
	average NP length	<i>avg_np_len</i>	<i>mean</i> (length of NPs)

Appendix 2. Collinearity matrices (by language and speaker type).







Appendix 3. Research repository on GitHub.

https://github.com/eaklykova/syntactic_complexity

Appendix 4. Python package for calculating syntactic complexity measures.

<https://pypi.org/project/syntaxcomp/>